



DFDOF

Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	ft-df019-2018-ios
Operator:	Floris
Analysis Started:	2026-06-18 11:45:45
Analysis Completed:	2026-06-18 11:56:13
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\test_mavic_pro\df019\2018\ios
Output Directory:	C:\Users\Floris\Documents\dji_drones\test_mavic_pro\df019\2018\dfdof_output\ft-df019-2018-ios
Evidence Sources:	3
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner. The populated case output directory is the authoritative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
df019_internal.001	drone_flight_storage	physical	3.7 GB	2026-06-18 11:45:58
internal_dji_export.zip	drone_flight_storage	logical	177.1 MB	2026-06-18 11:45:59
ios_logical.zip	controller_ios	logical	372.3 MB	2026-06-18 11:46:00

Evidence Source Path

C:\Users\Floris\Documents\dji_drones\test_mavic_pro\df019\2018\ios

Cryptographic Hashes

df019_internal.001		drone_flight_storage	physical
SHA-256:	8aaa5ddc477d86c8a9448e79a0ca1529a3bef8b42d1f7414837d90c7f842b219		
SHA-1:	c1d5c772c5fc0c4d092244c86a8a4d6c55169908		
internal_dji_export.zip		drone_flight_storage	logical
SHA-256:	75fd20cfa8a9c81174a7dc23b98b0237b25745432ec26a1621974002556a48a3		
SHA-1:	b9c3ebf8eeb97a5a62ab3518c53d00e5231453af		
ios_logical.zip		controller_ios	logical
SHA-256:	7b9ebc8fd9c773ae02f7708d2f9720cd4764ebeb46141fa5598f84399d204361		
SHA-1:	9ead1ad6ccb5536a723c9332b518688edb8888b3		

Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p1 - drone flight storage]: mmls: no partition table found in df019_internal.001; treating as direct partition image
2. [p2 - controller android]: source evidence not found
3. [p3 - controller android]: source evidence not found
4. [p3 - controller ios - images]: no artefacts found
5. [p3 - controller ios - videos]: no artefacts found
6. [p3 - drone sd]: source evidence not found
7. [p4 - controller android]: source evidence not found
8. [p4 - drone sd]: source evidence not found
9. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY000.csv
10. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY004.csv
11. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY008.csv
12. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY009.csv
13. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY010.csv
14. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY011.csv
15. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY012.csv
16. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY013.csv
17. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY014.csv
18. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY015.csv
19. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY016.csv

Tool Execution Status

Tool	Invocations	Successes	Status
datcon	23	23	[OK]
fls	1	1	[OK]
icat	17	17	[OK]
mmls	1	0	[X] Failed
parser_ios	1	1	[OK]
txtlogtocsvtool	1	1	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	2 / 4
Artefact Categories With Data	5 / 7
Flights With Primary Correlation	1 / 12
Tools Succeeded	5 / 6

3. Device and Account Identification

Field	Value	Confidence	Sources
Installed DJI App(s)	com.dji.go	[K] Controller-only	1
Controller Device	iPad mini 4	[K] Controller-only	1
Drone Serial Number	08RDD9100100KG	[K] Controller-only	2
Drone Model	Mavic Pro MavicPro UNKNOWN	[I] Inconsistent	14
Controller Device	iPad5,1__weibosdk__003143000__iphone__os 11.4	[K] Controller-only	1
DJI Application Version	4.2.21	[K] Controller-only	2
Account E-Mail	Droneforensics@vtoinc.com	[K] Controller-only	1
Product Type	13	[K] Controller-only	1
Last Country Code	US	[K] Controller-only	1
Last Flight Date	1970-01-01T00:00:00Z	[K] Controller-only	1
Last Known Latitude (Find Aircraft)	39.96117417	[K] Controller-only	1
Last Known Longitude (Find Aircraft)	-106.21650423	[K] Controller-only	1
Last Known Altitude (Find Aircraft)	2.1	[K] Controller-only	1
Last Known Heading (Find Aircraft)	75.2	[K] Controller-only	1
Last Known Horiz. Accuracy (Find Aircraft)	1.0	[K] Controller-only	1

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: db063f013eb8a83d18b645d75d79a5c682617f9c7c66f961dde64b7842ee8b9d
Drone Log Name: FLY001.csv
Start: 2017-08-26T16:43:07.807000+00:00
End: 2017-08-26T16:55:14.777000+00:00
Duration: 727.0 s

Flight Metrics

Peak Height: -
Distance Travelled: -
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

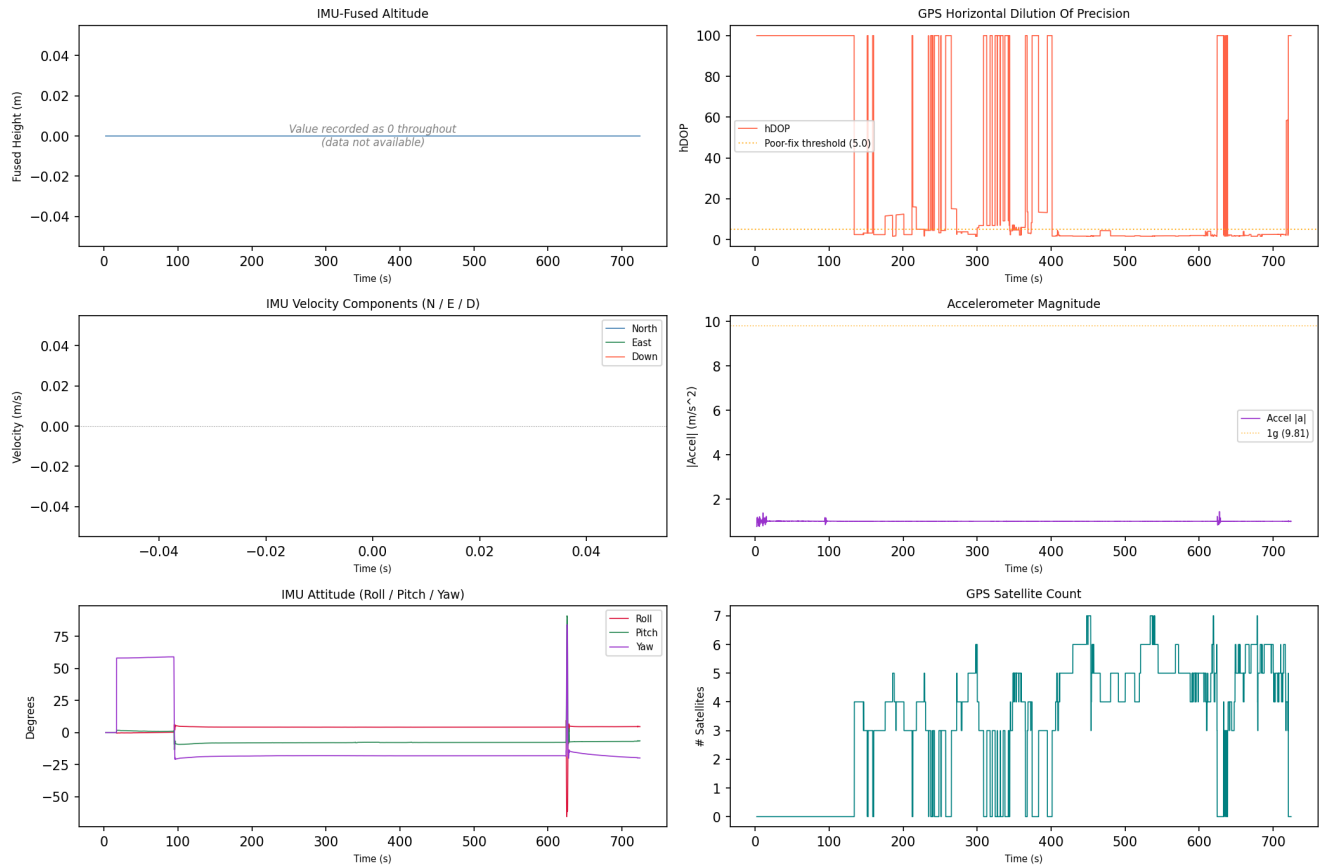
1. Manual
2. Atti
3. GPS_Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Reached peak height	-0.0 m
-	drone_flight_storage:dl	Motor turned off	off
2017-08-26 16:45:30	drone_flight_storage:dl	Log started	MavicPro
2017-08-26 16:55:14	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_01



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 02

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 7fdcf658562ef5ce293793a9e6e2b4df3f78eabc0f5e9e6324bd45e7e09cf4b4
Drone Log Name: FLY002.csv
Start: 2017-08-26T16:55:30.386000+00:00
End: 2017-08-26T17:02:33.406000+00:00
Duration: 423.0 s

Flight Metrics

Peak Height: -
Distance Travelled: -
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

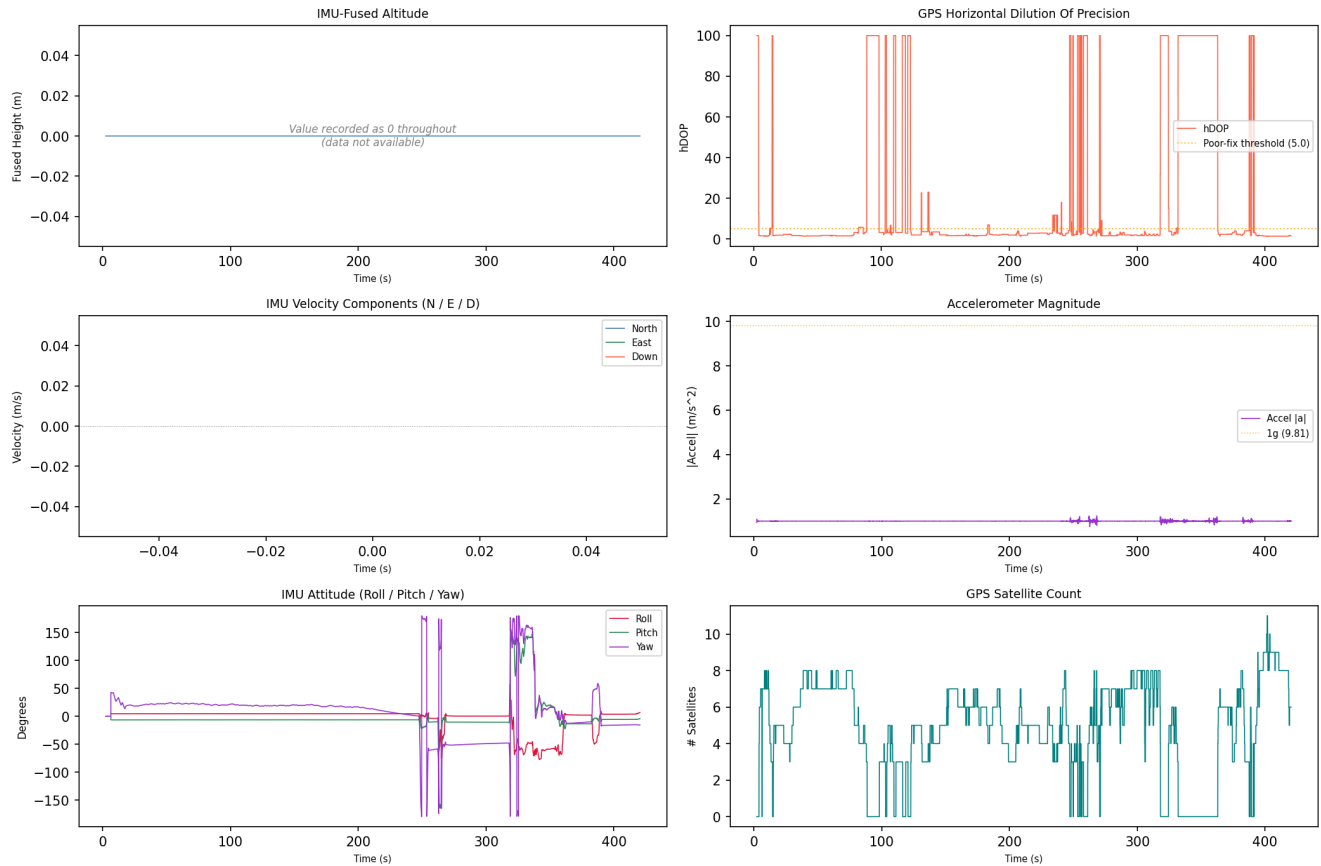
1. Manual
2. Atti
3. GPS_Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Reached peak height	-0.0 m
-	drone_flight_storage:dl	Motor turned off	off
2017-08-26 16:55:41	drone_flight_storage:dl	Log started	MavicPro
2017-08-26 17:02:34	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_02



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 03

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 9d91f72db0553b6dd6884a293a2aab5c1fc2ebe6afadb4049efcb8a4201c6586
Drone Log Name: FLY003.csv
Start: 2017-08-26T17:02:52.342000+00:00
End: 2017-08-26T17:10:56.252000+00:00
Duration: 483.9 s

Flight Metrics

Peak Height: -
Distance Travelled: -
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

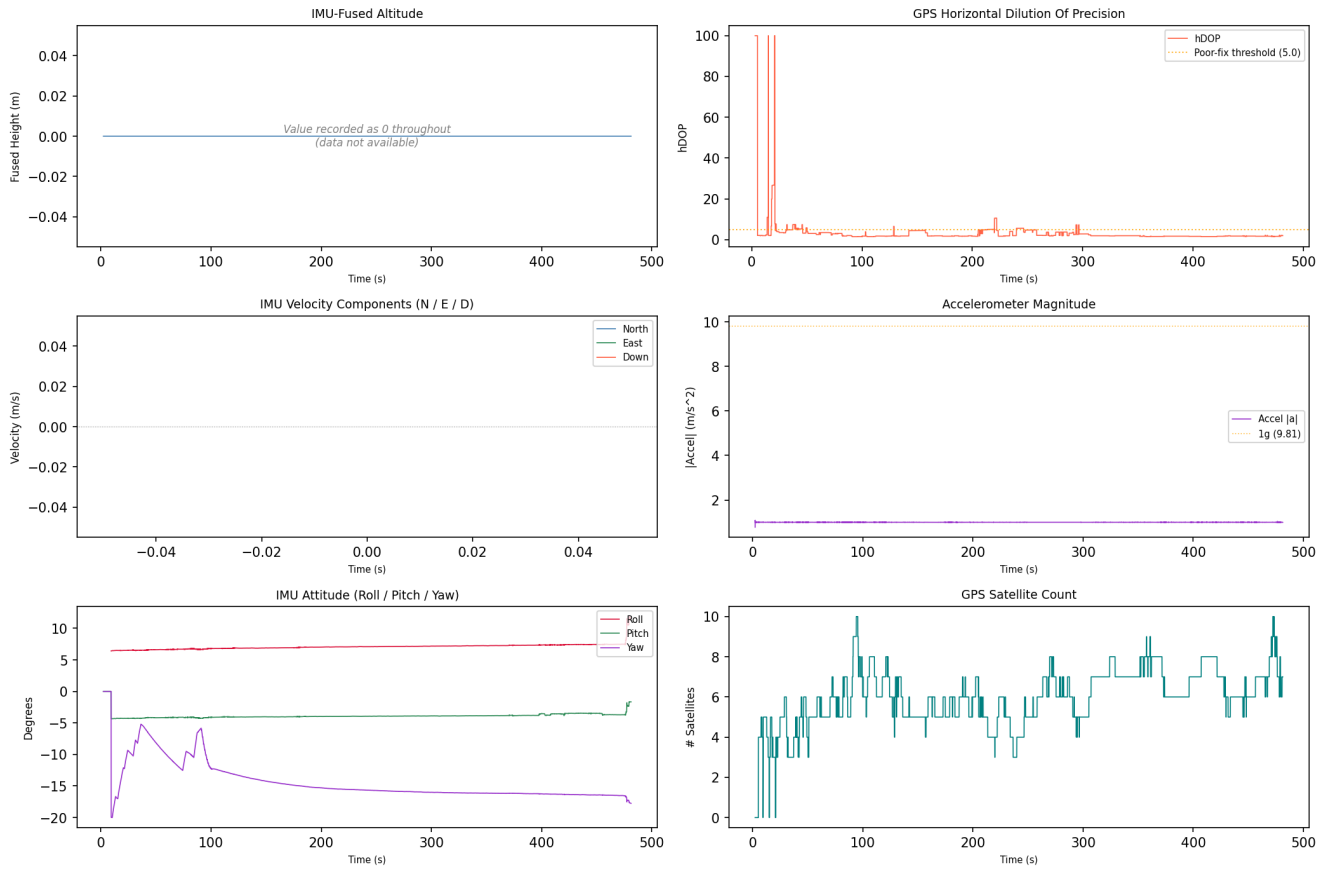
1. Manual
2. Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Reached peak height	-0.0 m
-	drone_flight_storage:dl	Motor turned off	off
2017-08-26 17:03:15	drone_flight_storage:dl	Log started	MavicPro
2017-08-26 17:10:56	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_03



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 04

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 60f006849b1314d076bde4059a2ab8addc09c5d4c2b285475f3435a6cfccd08c
Drone Log Name: FLY005.csv
Start: 2017-08-26T17:49:51.095000+00:00
End: 2017-08-26T18:00:03.898000+00:00
Duration: 612.8 s

Flight Metrics

Peak Height: -
Distance Travelled: -
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

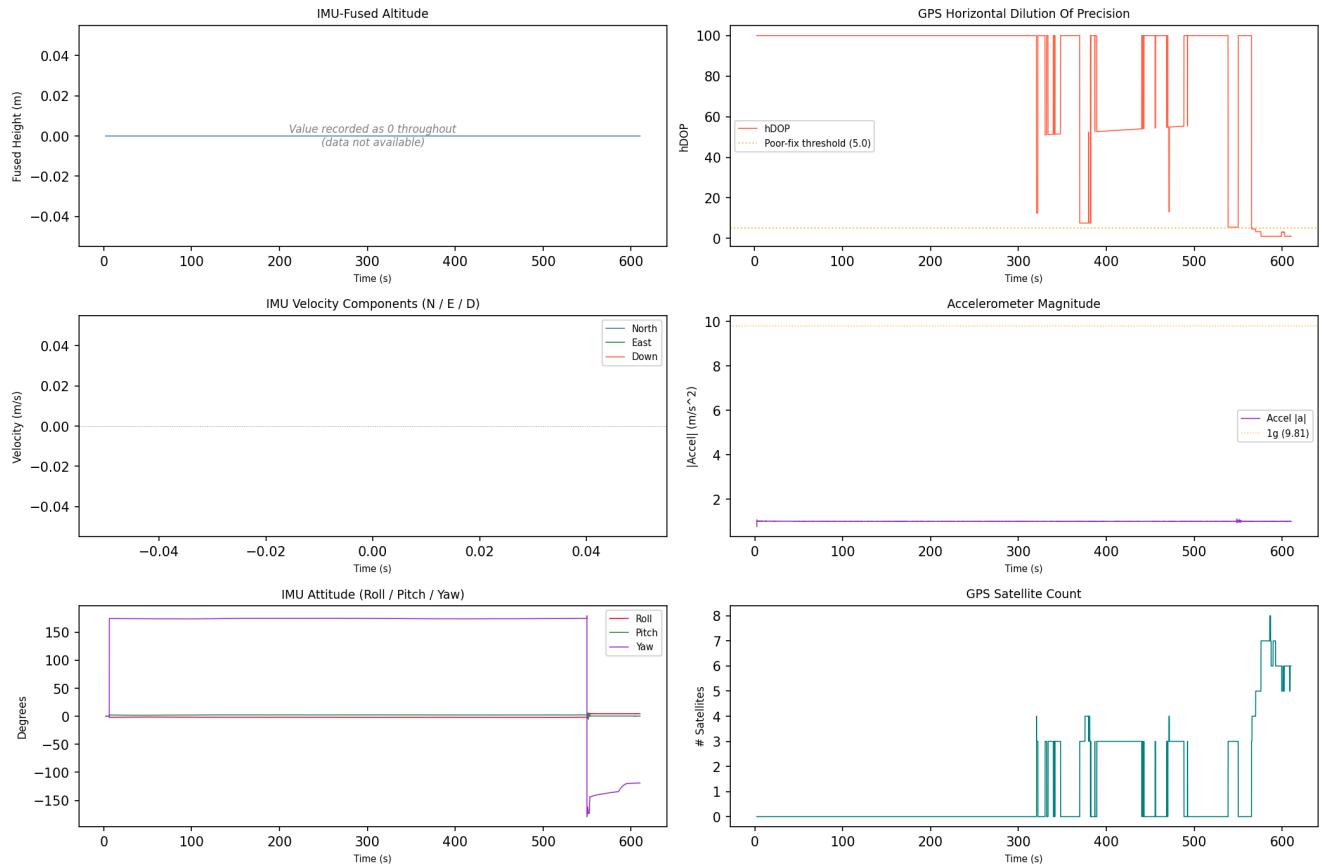
1. Manual
2. Atti
3. GPS_Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Reached peak height	-0.0 m
-	drone_flight_storage:dl	Motor turned off	off
2017-08-26 17:59:29	drone_flight_storage:dl	Log started	MavicPro
2017-08-26 18:00:04	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_04



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 05

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 858eb36ba0e57bd5a4148d7001b95494d7a5d955ae791ede82f560f67ac16e6c
Drone Log Name: FLY006.csv
Start: 2017-08-29T17:45:57.658000+00:00
End: 2017-08-29T17:55:13.138000+00:00
Duration: 555.5 s

Flight Metrics

Peak Height: 29.384265934076147 m
Distance Travelled: 115.51727098160579 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

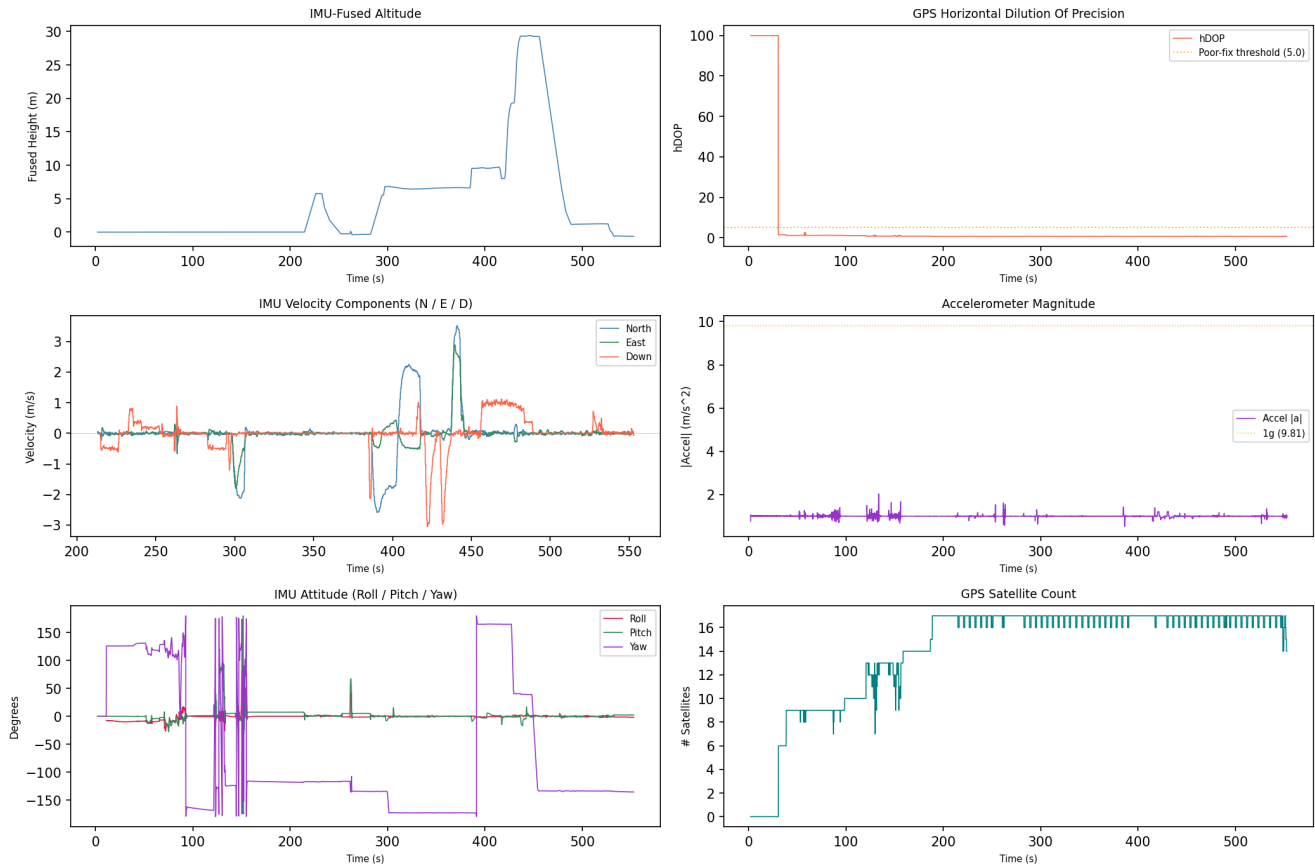
1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AutoTakeoff
6. AutoLanding
7. GoHome
8. Joystick
9. NaviSubMode_Pointing
10. FORCE_LANDING

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2017-08-29 17:46:31	drone_flight_storage:dl	Log started	MavicPro
2017-08-29 17:49:32	drone_flight_storage:dl	Motor turned on	on
2017-08-29 17:50:12	drone_flight_storage:dl	Motor turned off	off
2017-08-29 17:50:39	drone_flight_storage:dl	Motor turned on	on
2017-08-29 17:53:26	drone_flight_storage:dl	Reached peak height	29.4 m
2017-08-29 17:54:52	drone_flight_storage:dl	Motor turned off	off
2017-08-29 17:55:12	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

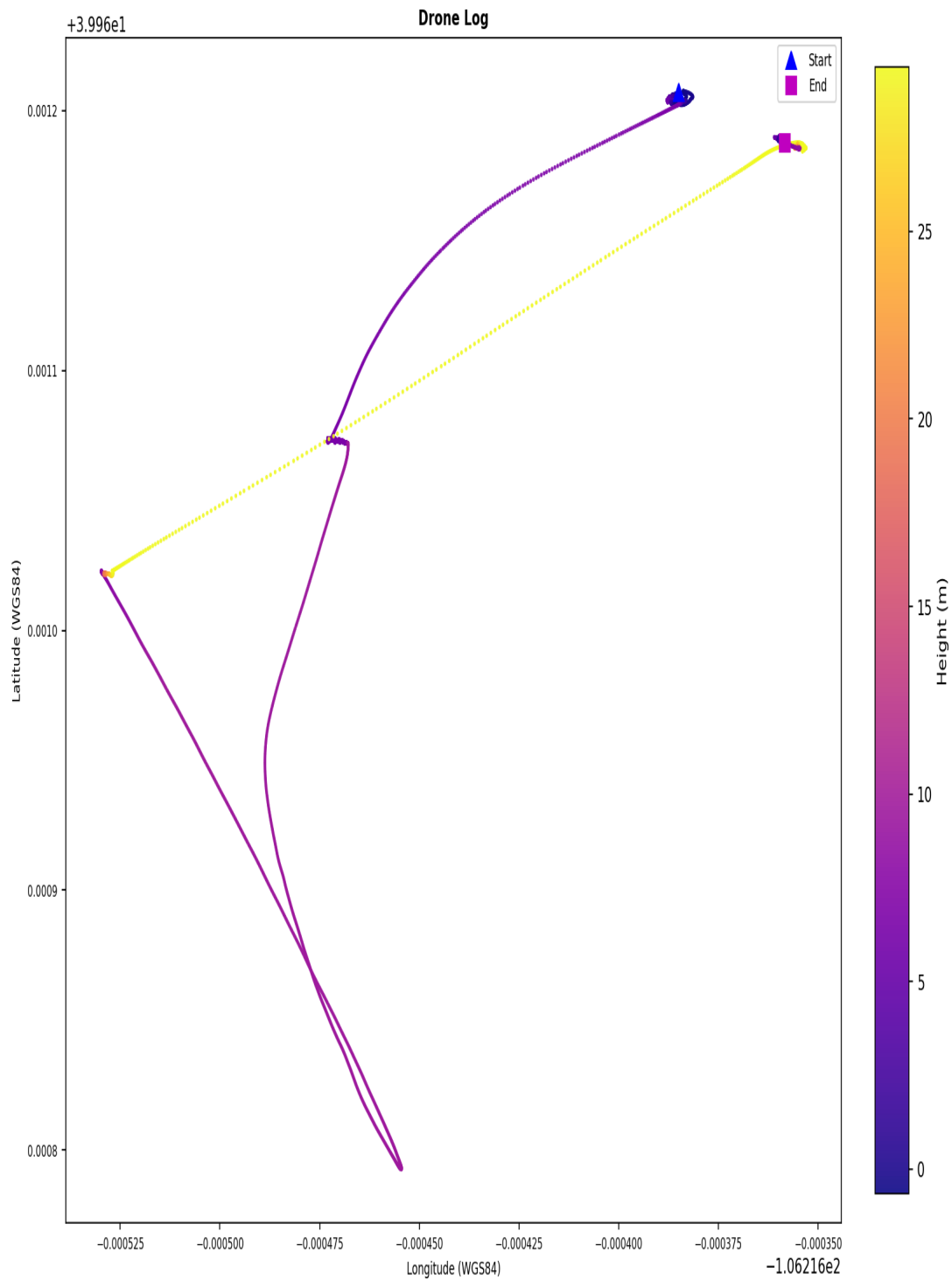
Drone Log Telemetry - flight_05



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_05



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY006.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY006.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 06

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 71b75b361507348338b9cdc089a2cc46c2501e331c08343e8b6c4cd3f44a96ce
Drone Log Name: FLY007.csv
Start: 2017-08-29T17:55:18.952000+00:00
End: 2017-08-29T18:14:56.883000+00:00
Duration: 1177.9 s

Flight Metrics

Peak Height: 28.472565241587052 m
Distance Travelled: 312.1688819870548 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

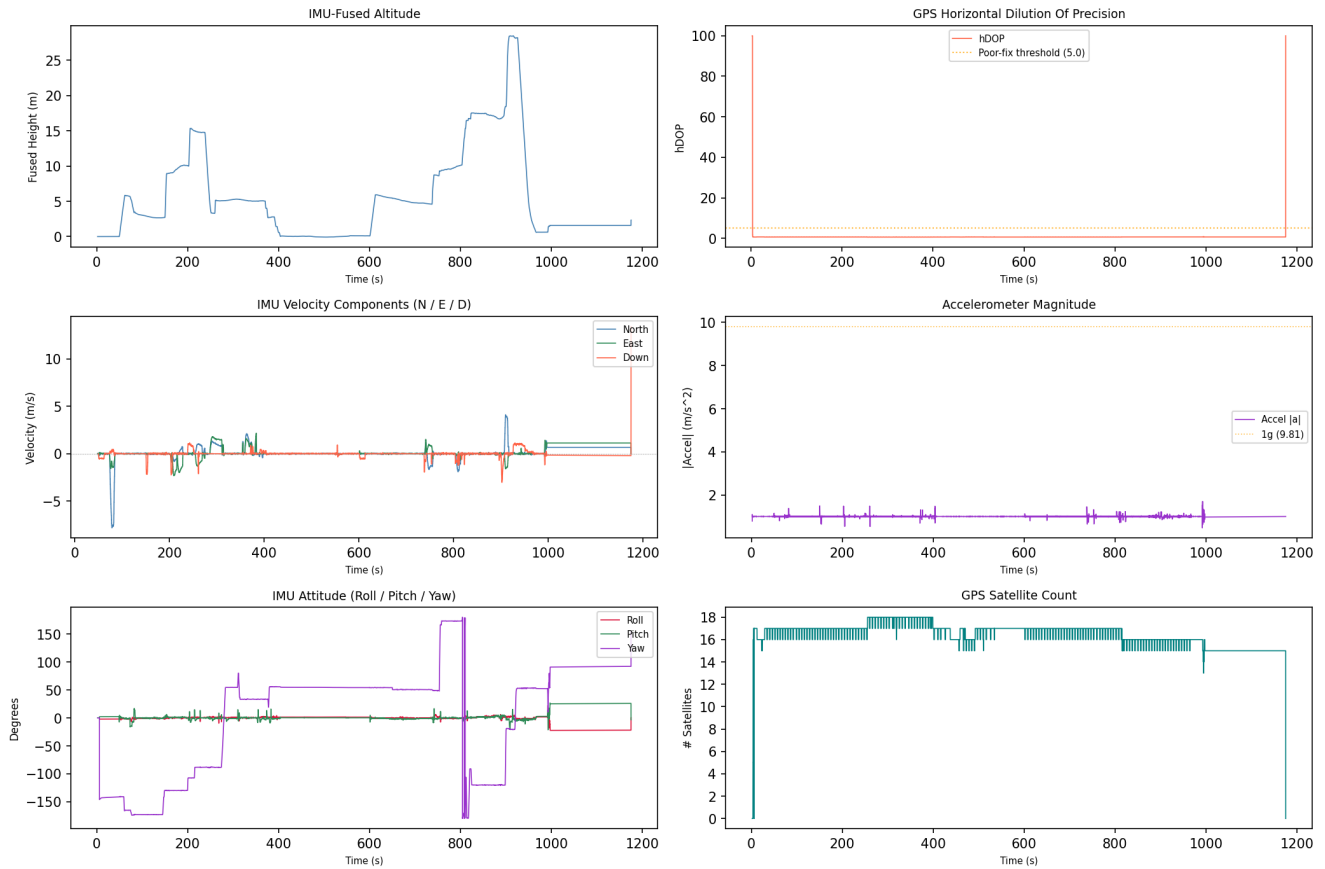
1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AutoTakeoff
6. Joystick
7. NaviSubMode_Pointing
8. AutoLanding
9. FORCE_LANDING
10. GoHome

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2017-08-29 17:55:26	drone_flight_storage:dl	Log started	MavicPro
2017-08-29 17:56:10	drone_flight_storage:dl	Motor turned on	on
2017-08-29 18:02:07	drone_flight_storage:dl	Motor turned off	off
2017-08-29 18:05:21	drone_flight_storage:dl	Motor turned on	on
2017-08-29 18:10:38	drone_flight_storage:dl	Reached peak height	28.5 m
2017-08-29 18:11:30	drone_flight_storage:dl	Motor turned off	off
2017-08-29 18:11:59	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_06



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

4. Flight Analysis - Flight 07

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: fcbfc5186935e537aa4ea7828e1f96bf90023d306f34ae61d8294768061a9f97
Drone Log Name: 2018-06-19_15-42-28_FLY057.csv
Start: 2018-06-19T21:40:55.374000+00:00
End: 2018-06-19T21:46:20.014000+00:00
Duration: 324.6 s

Flight Metrics

Peak Height: 0.6547713565293393 m
Distance Travelled: 4.614534115216808 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

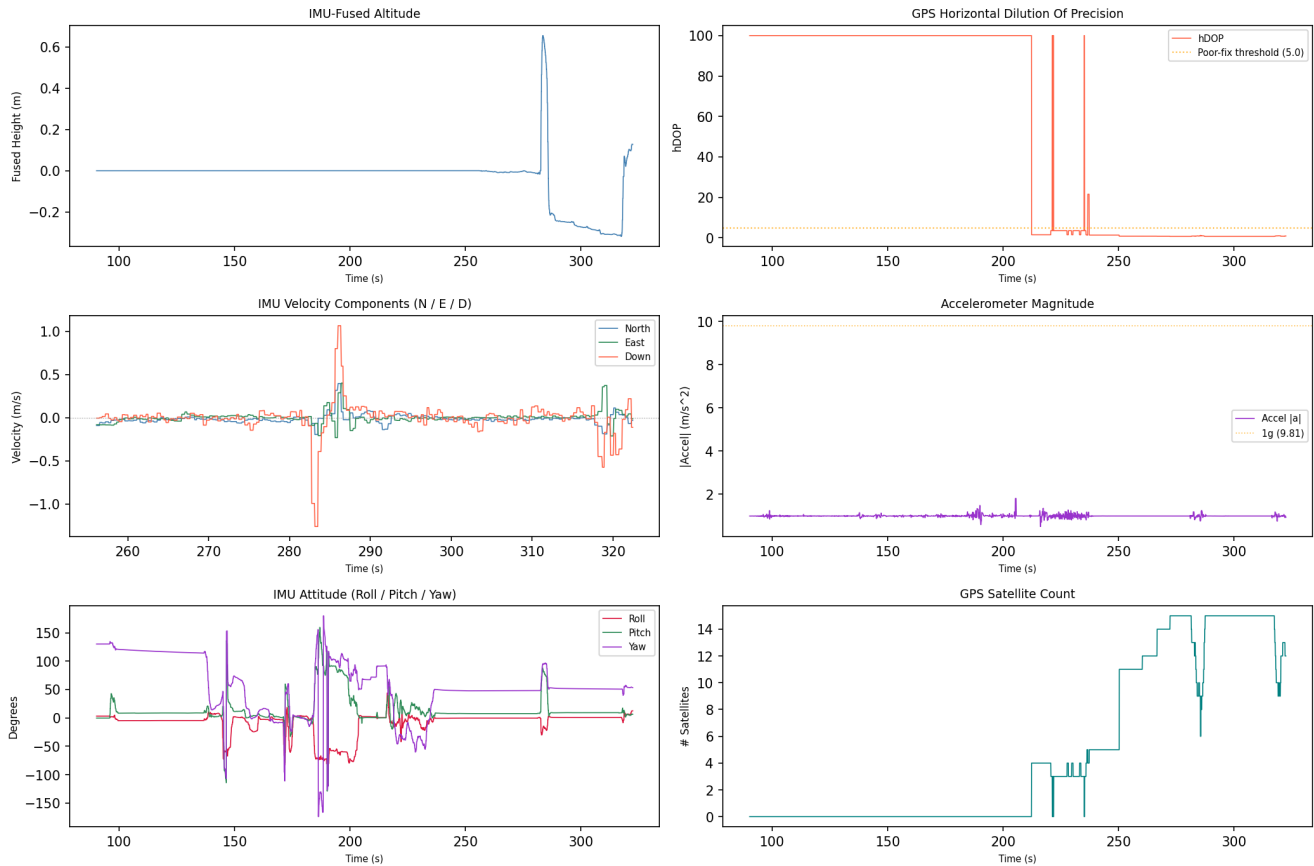
1. GPS_Atti
2. Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	ctrl_ios:dl	Motor turned off	off
2018-06-19 21:44:30	ctrl_ios:dl	Log started	MavicPro
2018-06-19 21:45:41	ctrl_ios:dl	Reached peak height	0.7 m
2018-06-19 21:46:20	ctrl_ios:dl	Log ended	

Drone Log - Sensor Telemetry

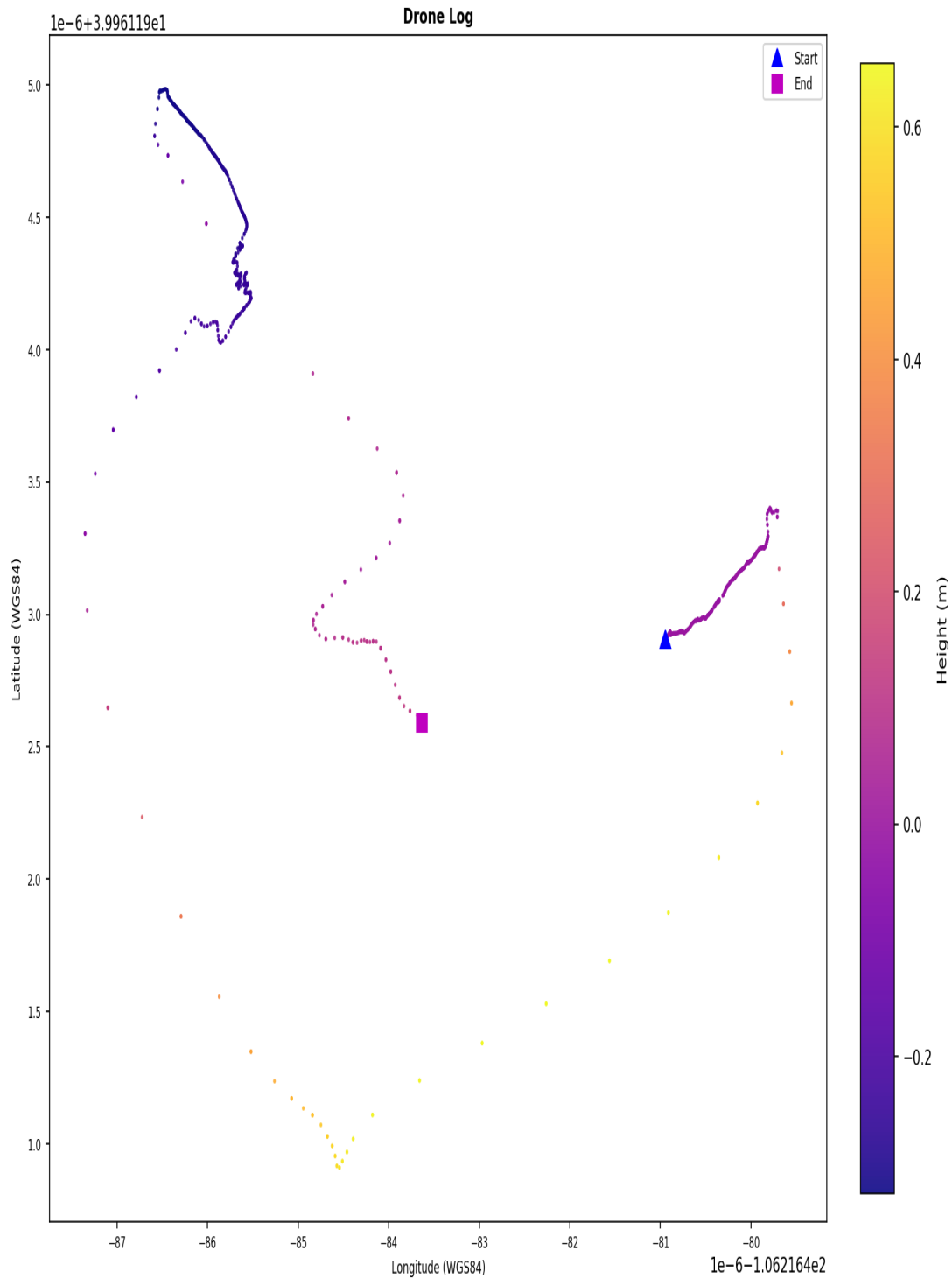
Drone Log Telemetry - flight_07



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_07



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: 2018-06-19_15-42-28_FLY057.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (2018-06-19_15-42-28_FLY057.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path.

Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 08

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 7d67d444d9a063e14c2f52a28f847a155a3ed52a9d54d9442f2bbca3993166b0
Drone Log Name: 2018-06-19_15-47-44_FLY058.csv
Start: 2018-06-19T21:46:29.097000+00:00
End: 2018-06-19T22:07:34.045000+00:00
Duration: 1264.9 s

Flight Metrics

Peak Height: 2.100362763564978 m
Distance Travelled: 45.73709757518451 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

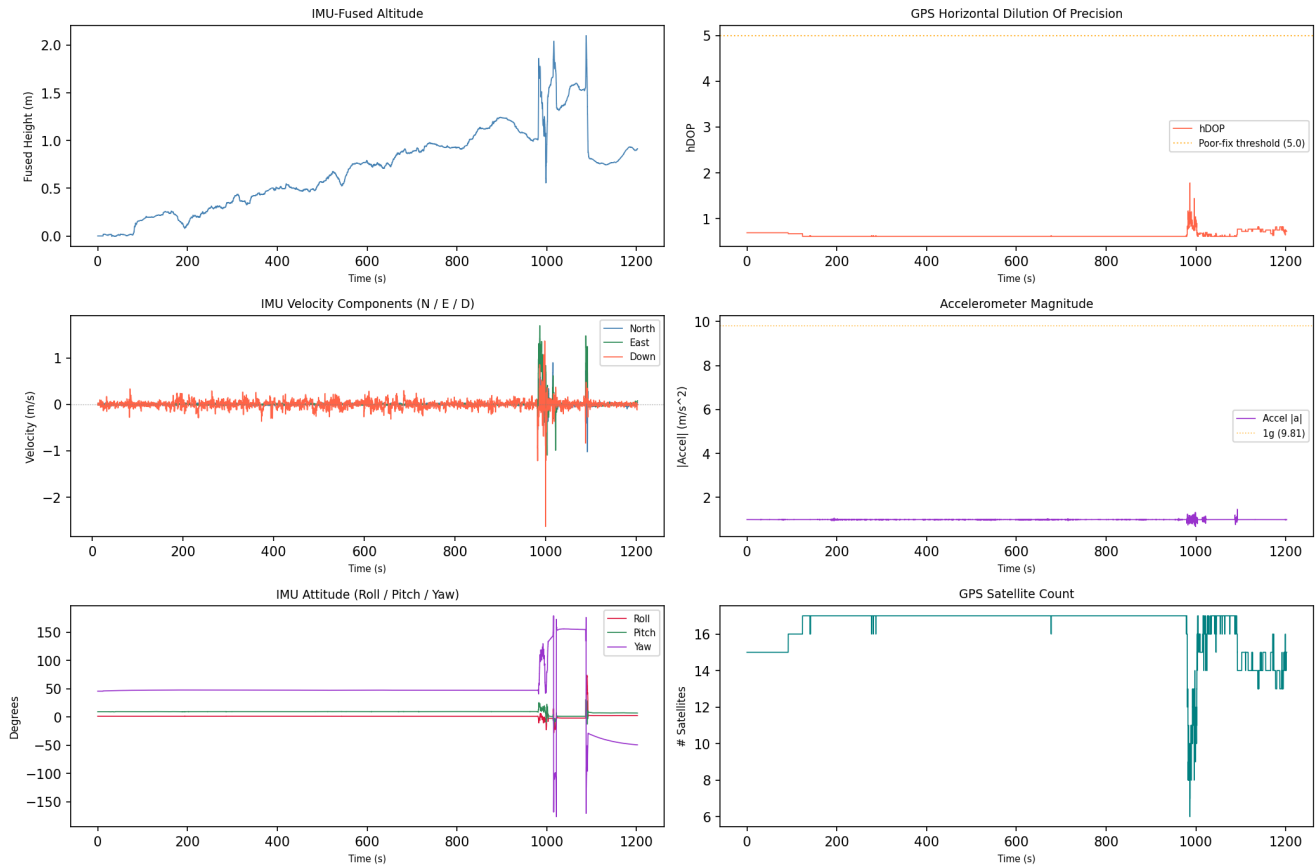
1. GPS_Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	ctrl_ios:dl	Motor turned off	off
2018-06-19 21:47:32	ctrl_ios:dl	Log started	MavicPro
2018-06-19 22:05:40	ctrl_ios:dl	Reached peak height	2.1 m
2018-06-19 22:07:34	ctrl_ios:dl	Log ended	

Drone Log - Sensor Telemetry

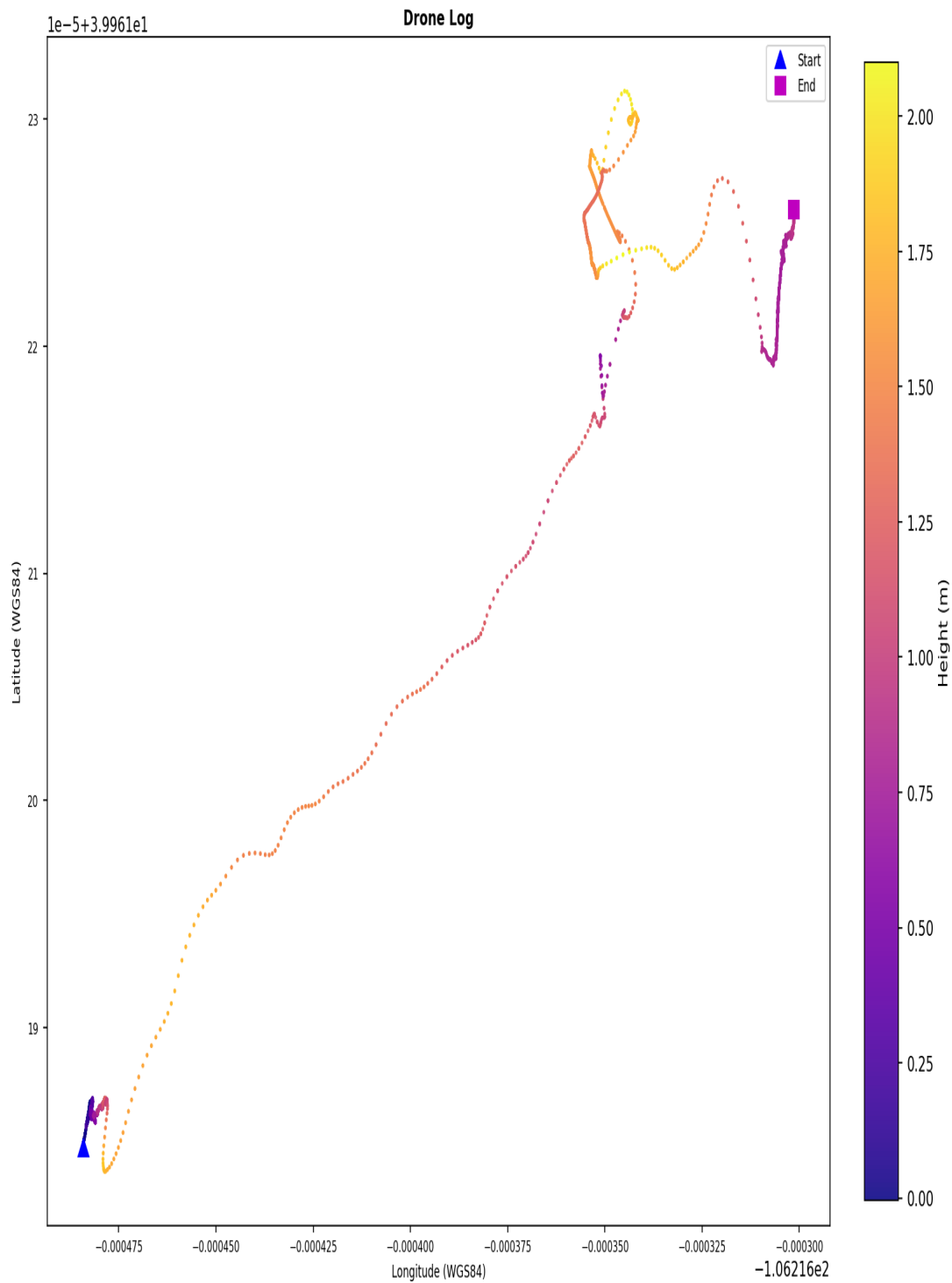
Drone Log Telemetry - flight_08



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_08



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: 2018-06-19_15-47-44_FLY058.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (2018-06-19_15-47-44_FLY058.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path.

Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 09

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: e267f904bcfc8a0aead9e88de72b31d1cdbe36f32f643aa7fe77f9077440742c
Drone Log Name: 2018-06-19_15-46-44_FLY058.csv
Start: 2018-06-19T21:46:29.265000+00:00
End: 2018-06-19T21:47:20.274000+00:00
Duration: 51.0 s

Flight Metrics

Peak Height: -
Distance Travelled: 0.9038390085446577 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

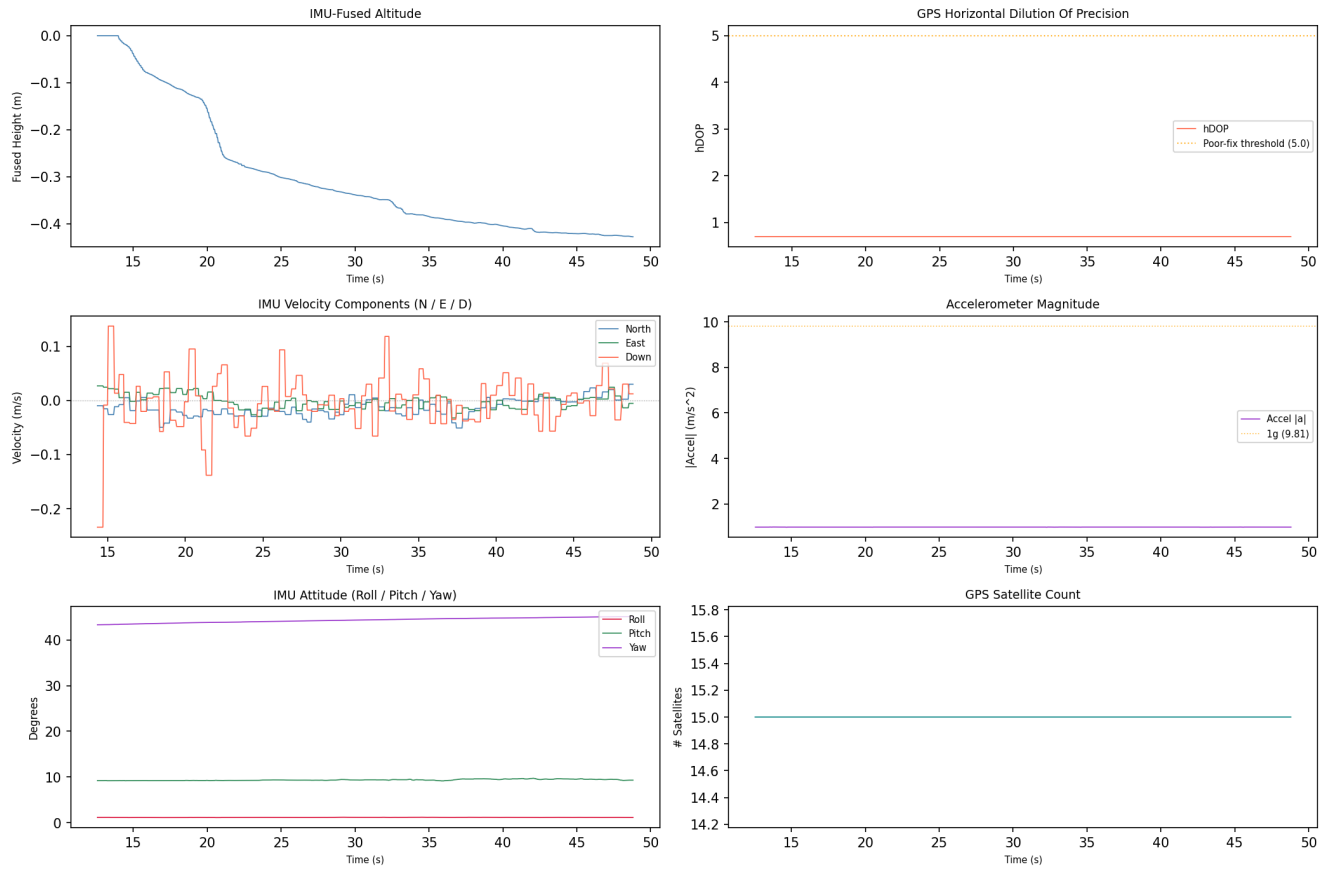
1. Atti
2. GPS_Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	ctrl_ios:dl	Motor turned off	off
2018-06-19 21:46:44	ctrl_ios:dl	Log started	MavicPro
2018-06-19 21:46:44	ctrl_ios:dl	Reached peak height	-0.0 m
2018-06-19 21:47:20	ctrl_ios:dl	Log ended	

Drone Log - Sensor Telemetry

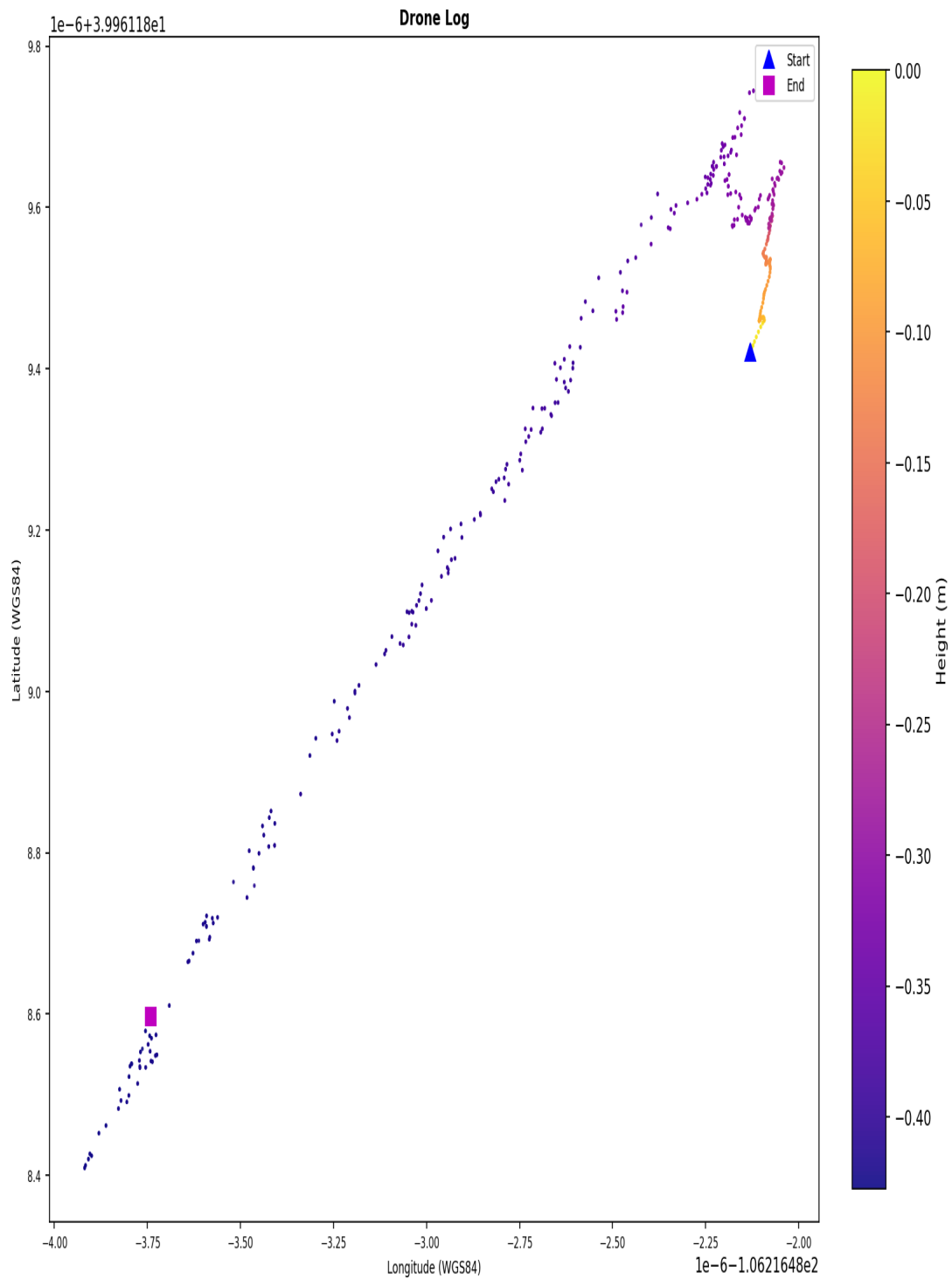
Drone Log Telemetry - flight_09



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_09



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: 2018-06-19_15-46-44_FLY058.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (2018-06-19_15-46-44_FLY058.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path.

Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 10

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 70ab9683610ae8e9c0f2e68ce9bd8e28345351fe0ee6839f962a59d1dc4b60d5
Drone Log Name: 2018-06-19_16-31-40_FLY059.csv
Start: 2018-06-19T22:31:26.381000+00:00
End: 2018-06-19T22:45:22.108000+00:00
Duration: 835.7 s

Flight Metrics

Peak Height: -
Distance Travelled: 26.289454136920327 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

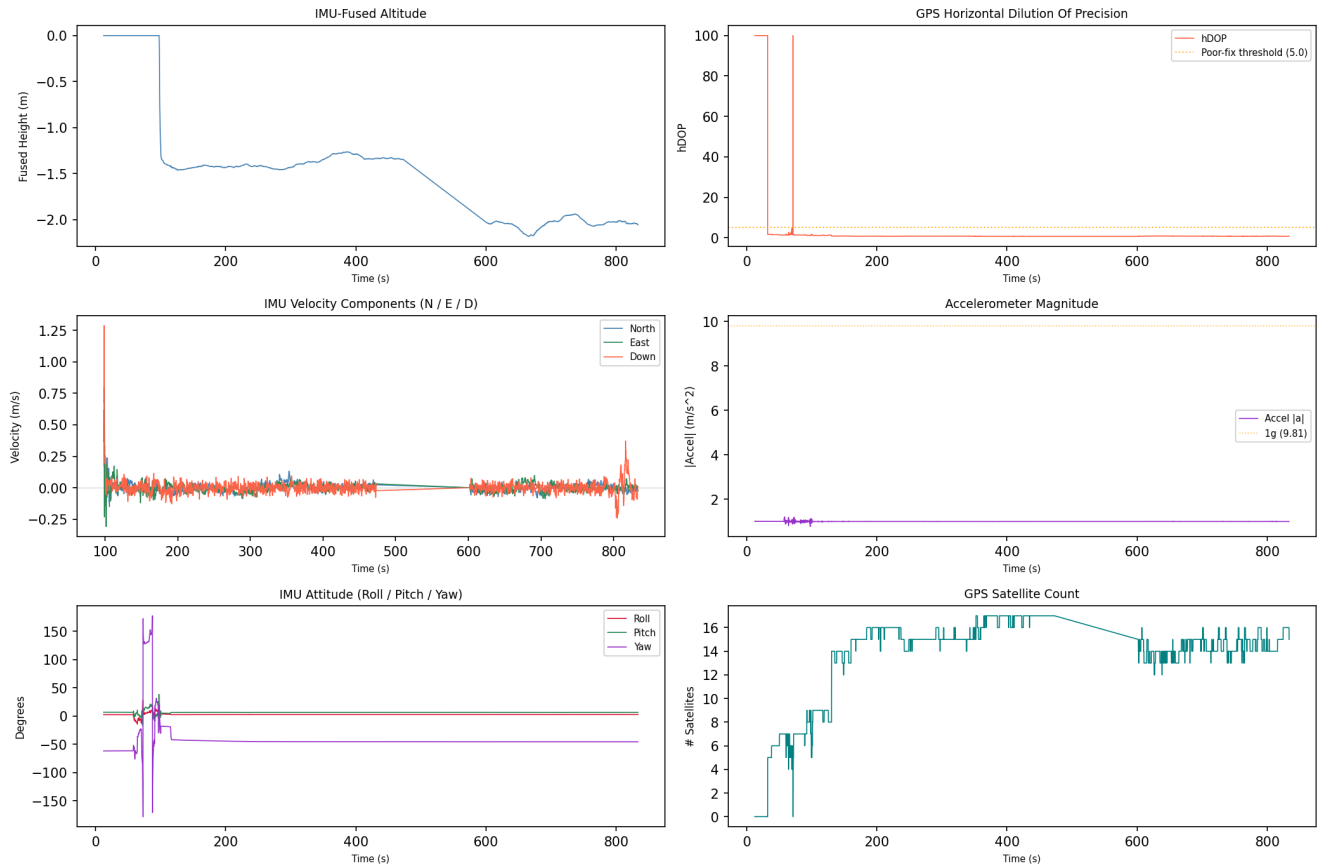
1. GPS_Atti
2. Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	ctrl_ios:dl	Reached peak height	-0.0 m
-	ctrl_ios:dl	Motor turned off	off
2018-06-19 22:32:01	ctrl_ios:dl	Log started	MavicPro
2018-06-19 22:45:21	ctrl_ios:dl	Log ended	

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_10



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

4. Flight Analysis - Flight 11

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: e14d09e21de6c3ac0a7e7887eeb0887921a713b9a08facf1c211fa9b5b00673f
Drone Log Name: 2018-06-19_16-45-50_FLY060.csv
Start: 2018-06-19T22:45:22.218000+00:00
End: 2018-06-19T22:46:50.310000+00:00
Duration: 88.1 s

Flight Metrics

Peak Height: -
Distance Travelled: 2.6815794565294557 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

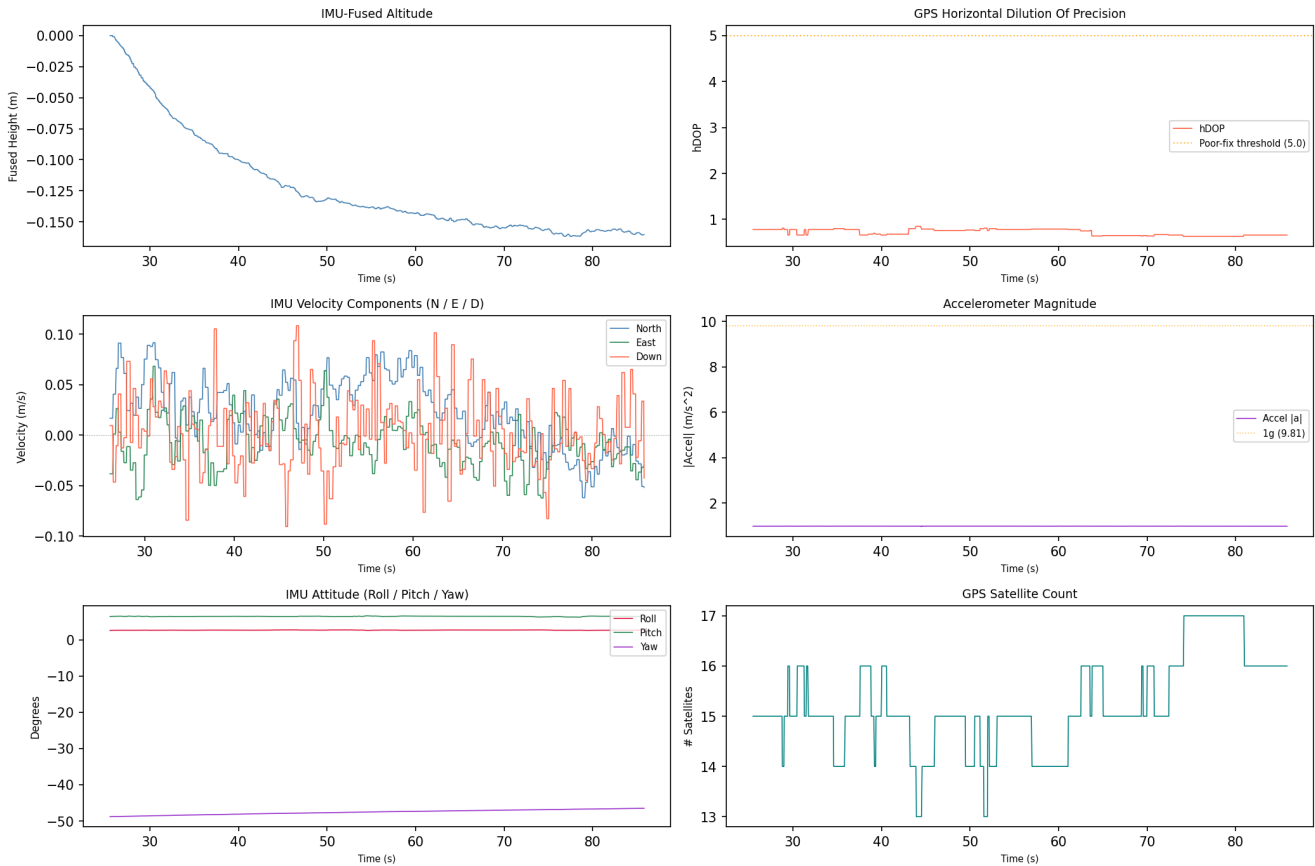
1. GPS_Atti

Key Events

Timestamp (UTC)	Source	Event	Details
2018-06-19 22:45:50	ctrl_ios:dl	Log started	UNKNOWN
2018-06-19 22:45:50	ctrl_ios:dl	Reached peak height	-0.0 m
2018-06-19 22:45:50	ctrl_ios:dl	Motor turned off	off
2018-06-19 22:46:49	ctrl_ios:dl	Log ended	

Drone Log - Sensor Telemetry

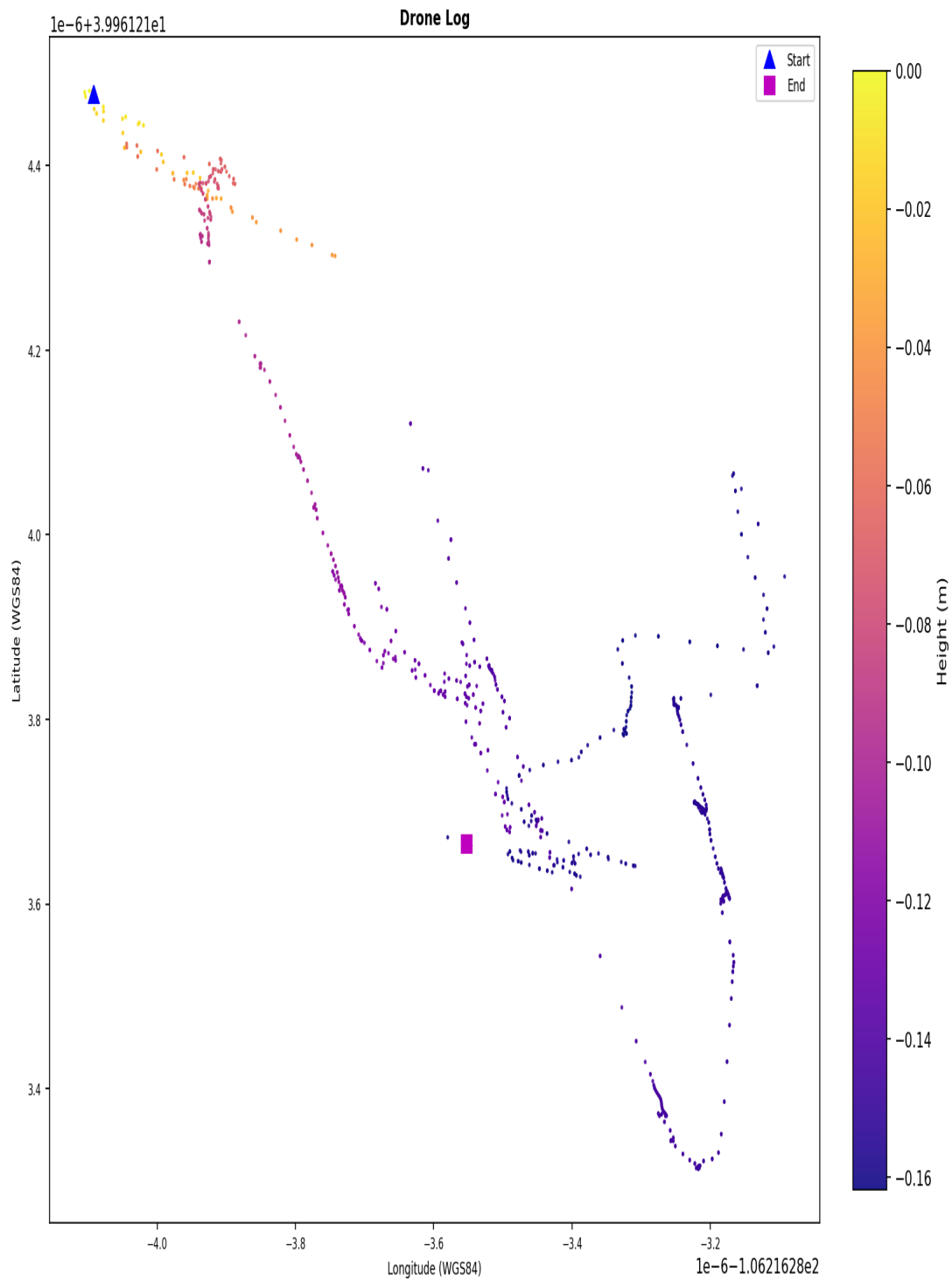
Drone Log Telemetry - flight_11



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_11



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: 2018-06-19_16-45-50_FLY060.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (2018-06-19_16-45-50_FLY060.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path.

Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 12

Correlation and Flight Window

Status:	Matched (primary GPS+temporal)
Confidence:	high
GPS Overlap:	497.8 s
Median GPS Distance:	0.14 m
Overlap:	100%
GPS Match Rate:	100%
Flight Record SHA-256:	91f805fb448f146a1bb3f4c6e4d0c4329da48a91b7c57a578630af87218d4094
Flight Record Name:	DJIFlightRecord_2018-06-19_17-00-38_.csv
Start:	2018-06-19T23:00:38.817000+00:00
End:	2018-06-19T23:08:56.618000+00:00
Duration:	497.8 s
Drone Log SHA-256:	ff6d76c2c297f62cfc78d1c17861e82854a316f326507d5a3fe911b3fa4f216c
Drone Log Name:	2018-06-19_16-56-14_FLY061.csv
Start:	2018-06-19T22:55:51.054000+00:00
End:	2018-06-19T23:09:09.700000+00:00
Duration:	798.6 s

Flight Metrics

Peak Height:	24.3167265436723 m
Distance Travelled:	1610.17 m
Video Recording:	No
Photo Capture:	No
Photos Taken:	0
Recording Duration:	0.0 s
Record Complete:	Yes

Fly Modes Observed (Chronological)

1. Atti
2. GPS_Atti
3. ASST_TAKEOFF
4. EngineStart
5. AutoTakeoff
6. FORCE_LANDING
7. ForceLanding

Key Events

Timestamp (UTC)	Source	Event	Details
-	ctrl_ios:dl	Motor turned off	off
2018-06-19 22:56:16	ctrl_ios:dl	Log started	MavicPro
2018-06-19 23:00:38	ctrl_ios:dl	Motor turned on	on
2018-06-19 23:00:38	ctrl_ios:fr (v4.2.21)	Log started	SN: 08RDD9100100KG; App: 4.2.21; Mavic Pro
2018-06-19 23:00:38	ctrl_ios:fr (v4.2.21)	Motor turned on	on
2018-06-19 23:00:38	ctrl_ios:fr (v4.2.21)	Record mode changed	No
2018-06-19 23:04:25	ctrl_ios:dl	Reached peak height	24.3 m
2018-06-19 23:04:38	ctrl_ios:fr (v4.2.21)	Reached peak height	26.3 m
2018-06-19 23:08:56	ctrl_ios:dl	Motor turned off	off
2018-06-19 23:08:56	ctrl_ios:fr (v4.2.21)	Log ended	Dist home: 1.5 m; Height: 0.0 m; Rec: 0.0 s
2018-06-19 23:08:56	ctrl_ios:fr (v4.2.21)	Motor turned off	off
2018-06-19 23:09:09	ctrl_ios:dl	Log ended	

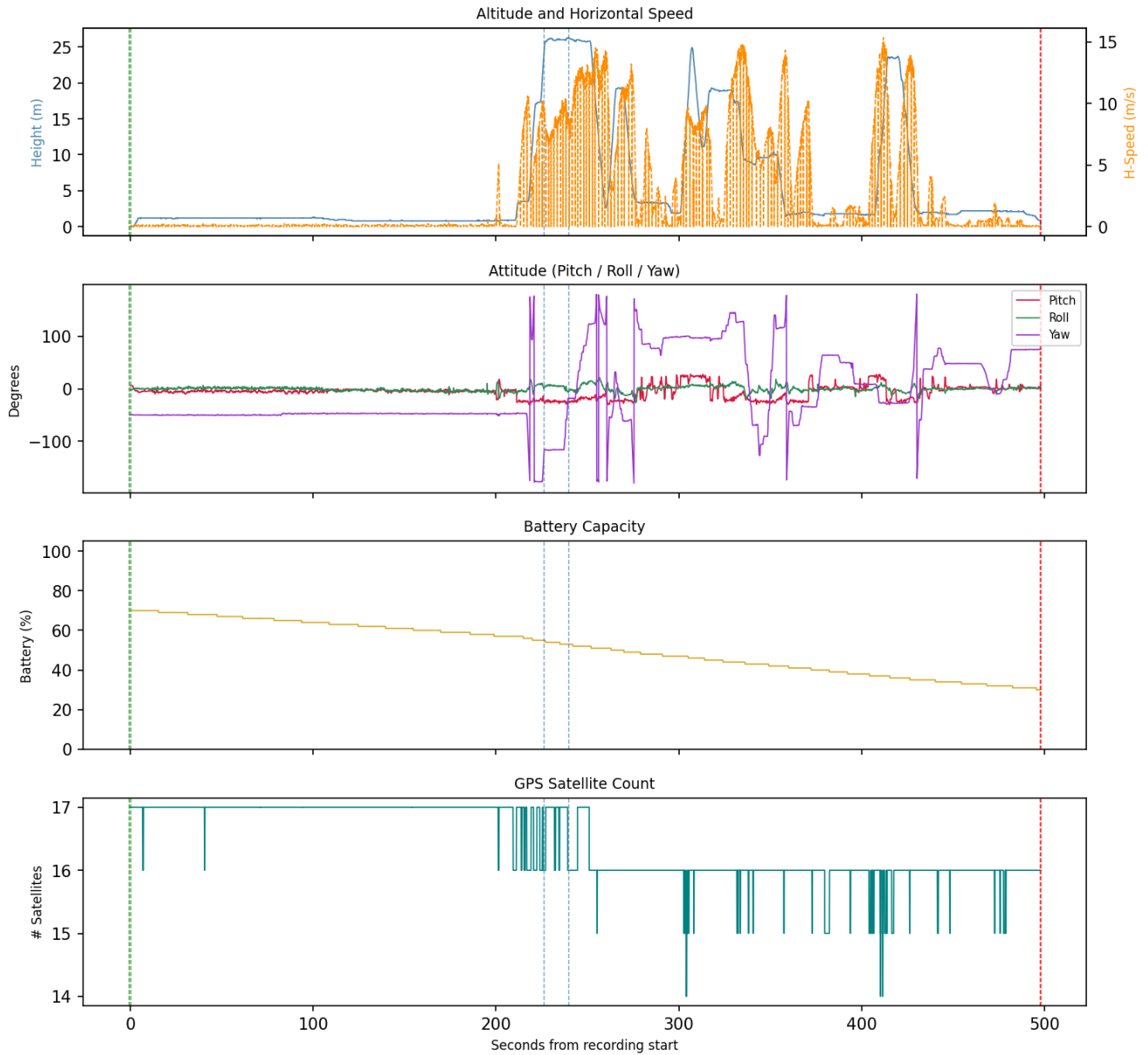
Correlated Artefacts

Possibly correlated (indirect indicators):

controller_ios: flight_logs	Log messages match (6 entries, format: json_array_log)
p5:dc6ea1c3002ee63fbfb3cc3bb23389853a040647beea0ab3622a0d0c915af2ac	

Flight Record - Sensor Telemetry

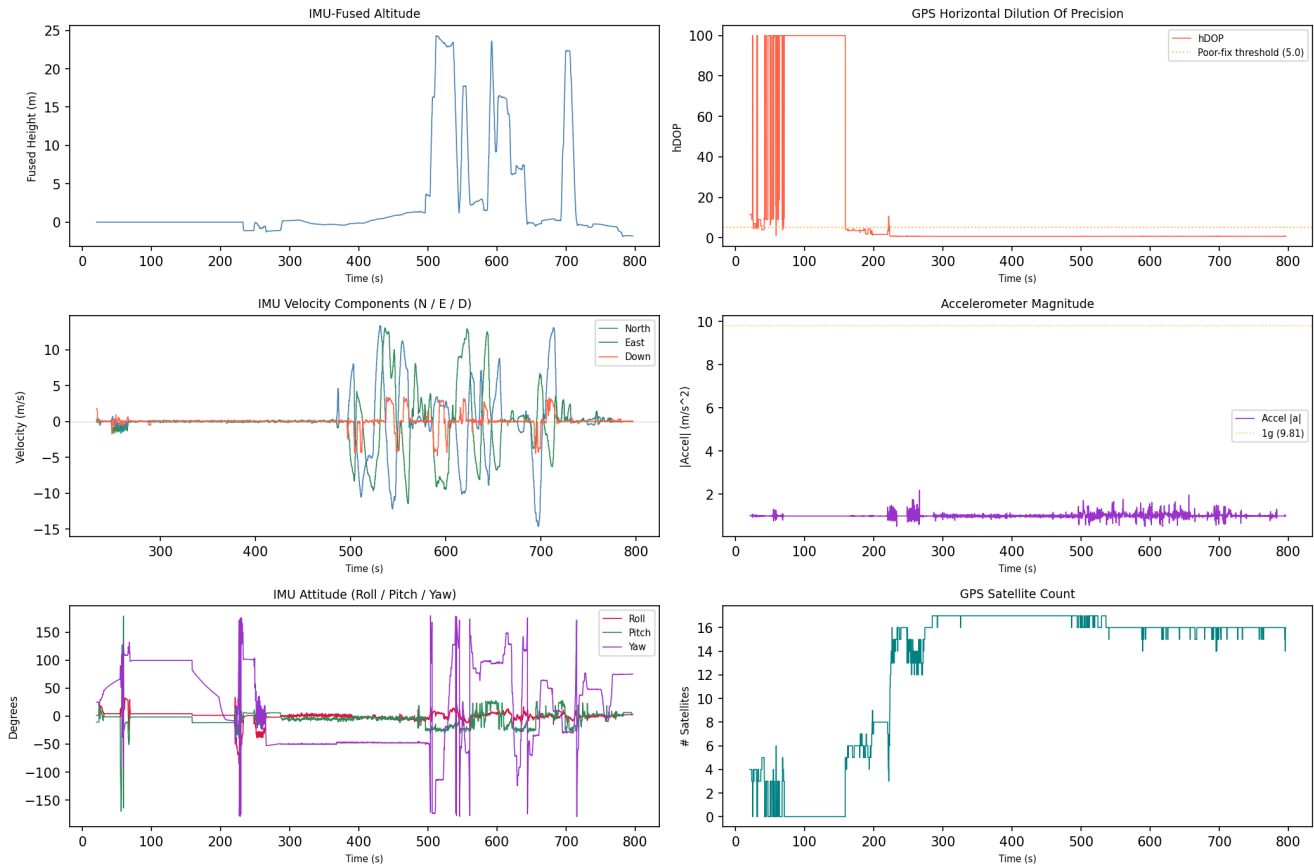
Flight Record Telemetry - flight_12



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

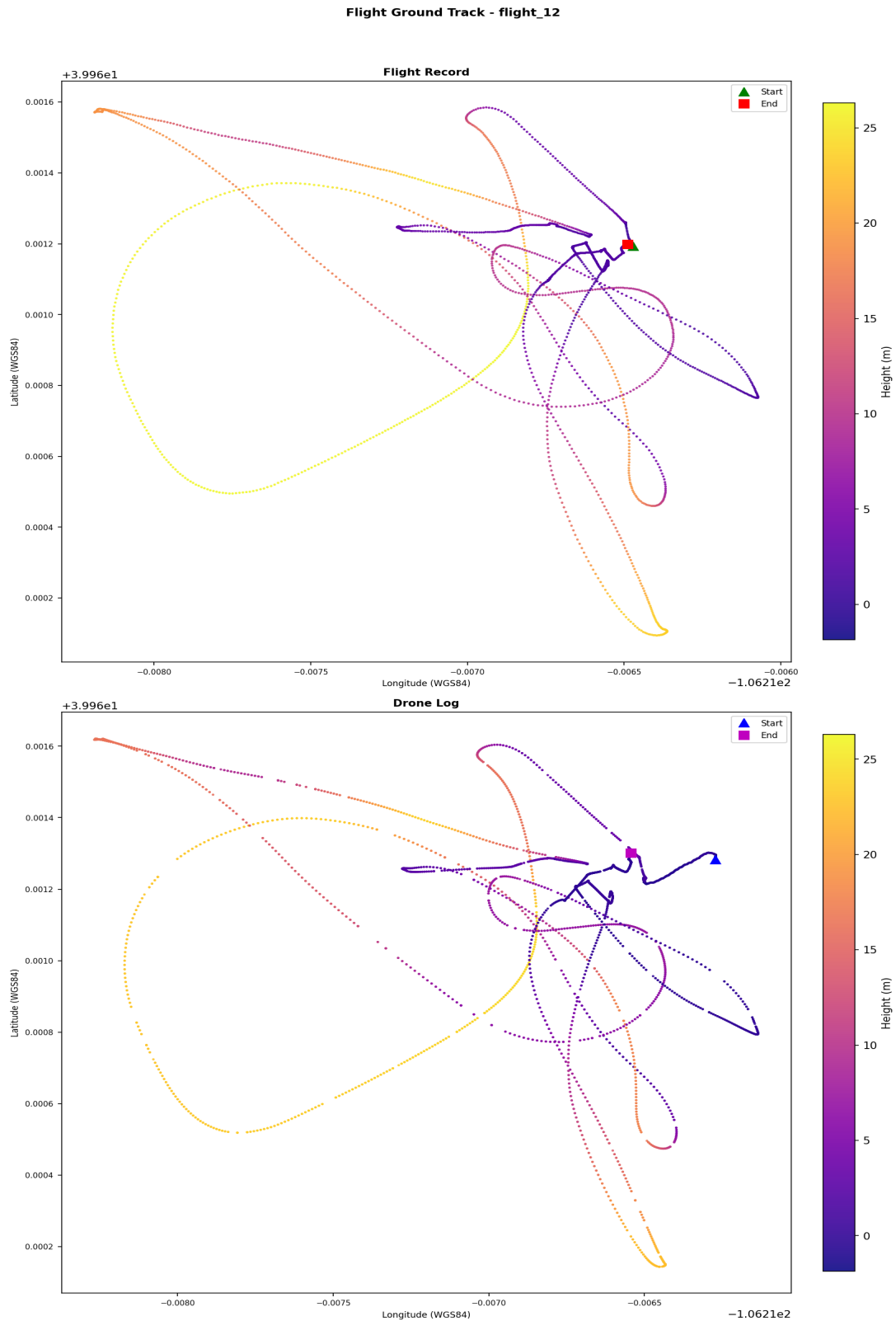
Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_12



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (2018-06-19_16-56-14_FLY061.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path.

Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	No
controller_ios	Yes
drone_sd	No
drone_flight_storage	Yes

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_ios	1	10	6	3	1	0	0
drone_flight_storage	0	0	17	0	0	0	0

Data Quality Notes per Category

Databases (10 files):

- 3 database(s) empty (controller_ios)

Drone Logs (23 files):

- 23 file(s) with row anomalies
- 23 file(s) with empty columns
- 23 file(s) with constant-value columns

Flight Logs (3 files):

- 1 file(s) unreadable (binary)
- formats: 1 bracketed_log, 1 json_array_log

Flight Records (1 files):

- 1 file(s) with empty columns
- 1 file(s) with constant-value columns

6. Data Quality and Anomalies

Data Quality Observations

- Drone_logs: 23 file(s) with row anomalies detected in telemetry data.
- Drone_logs: 23 file(s) with empty columns - some telemetry channels contain no data.
- Drone_logs: 23 file(s) with constant-value columns - some telemetry channels may be unreliable.
- Flight_logs: 1 file(s) unreadable (binary) - file content could not be parsed.
- Flight_logs: log files present in the following format(s): 1 bracketed_log, 1 json_array_log.
- Flight_records: 1 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 1 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

1. [p1 - drone flight storage]: mmls: no partition table found in df019_internal.001; treating as direct partition image
2. [p2 - controller android]: source evidence not found
3. [p3 - controller android]: source evidence not found
4. [p3 - controller ios - images]: no artefacts found
5. [p3 - controller ios - videos]: no artefacts found
6. [p3 - drone sd]: source evidence not found
7. [p4 - controller android]: source evidence not found
8. [p4 - drone sd]: source evidence not found
9. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY000.csv
10. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY004.csv
11. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY008.csv
12. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY009.csv
13. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY010.csv
14. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY011.csv
15. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY012.csv
16. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY013.csv
17. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY014.csv
18. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY015.csv
19. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY016.csv

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Databases

5255DF85724869CD36D4AFCD12AE092D.db

- database_empty: all tables empty

collectionList.db

- database_empty: all tables empty

djiFMDB.db

- database_empty: all tables empty

Drone Logs

norm_2018-06-19_15-42-28_FLY057.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 6 row(s) affected
- zero_coordinate: 1498 row(s) affected
- altitude_negative: 494 row(s) affected
- gps_poor_fix: 1801 row(s) affected
- contains_no_value: 4 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C
- contains_constant_value: 67 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, BatteryInfo:dyna_cnt:D, BatteryInfo:f_cap:D, BatteryInfo:l_cell:D (+62 more)
- gps_accuracy_poor: 1616 row(s) affected

- attitude_out_of_bounds: 83 row(s) affected

norm_2018-06-19_15-46-44_FLY058.csv

- missing_gps: 7 row(s) affected
- altitude_negative: 632 row(s) affected
- contains_no_value: 7 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, IMU_ATT(0):relativeHeight:C (+2 more)
- contains_constant_value: 83 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, BatteryInfo:cap_per:D, BatteryInfo:dyna_cnt:D, BatteryInfo:f_cap:D (+78 more)

norm_2018-06-19_15-47-44_FLY058.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 2 row(s) affected
- motor_airborne_off: 5178 row(s) affected
- contains_no_value: 4 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C
- contains_constant_value: 68 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, BatteryInfo:dyna_cnt:D, BatteryInfo:f_cap:D, BatteryInfo:out_ctl:D (+63 more)

norm_2018-06-19_16-31-40_FLY059.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 9 row(s) affected
- zero_coordinate: 155 row(s) affected
- altitude_negative: 9454 row(s) affected
- gps_poor_fix: 164 row(s) affected
- contains_no_value: 4 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C
- contains_constant_value: 65 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, BatteryInfo:dyna_cnt:D, BatteryInfo:f_cap:D, BatteryInfo:out_ctl:D (+60 more)
- gps_accuracy_poor: 159 row(s) affected

norm_2018-06-19_16-45-50_FLY060.csv

- missing_gps: 7 row(s) affected
- altitude_negative: 806 row(s) affected
- contains_no_value: 7 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, IMU_ATT(0):relativeHeight:C (+2 more)
- contains_constant_value: 80 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, BatteryInfo:dyna_cnt:D, BatteryInfo:f_cap:D, BatteryInfo:l_cell:D (+75 more)

norm_2018-06-19_16-56-14_FLY061.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 8 row(s) affected
- altitude_negative: 4131 row(s) affected
- altitude_spike: 7 row(s) affected
- gps_poor_fix: 623 row(s) affected
- contains_no_value: 4 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C
- contains_constant_value: 42 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D, BatteryInfo:out_ctl_f:D (+37 more)
- gps_accuracy_poor: 758 row(s) affected
- attitude_out_of_bounds: 39 row(s) affected

norm_FLY000.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 5 row(s) affected
- zero_coordinate: 4156 row(s) affected
- gps_poor_fix: 4156 row(s) affected
- contains_no_value: 24 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+19 more)
- contains_constant_value: 114 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+109 more)
- gps_accuracy_poor: 4156 row(s) affected
- attitude_out_of_bounds: 754 row(s) affected

norm_FLY001.csv

- missing_gps: 11 row(s) affected

- zero_coordinate: 1388 row(s) affected
- gps_poor_fix: 9531 row(s) affected
- contains_no_value: 18 column(s) - 4.3.0, ConvertDatV3, IMUCalcs(0):Lat:C, IMUCalcs(0):Long:C, IMUCalcs(0):diffVelD:C (+13 more)
- contains_constant_value: 93 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+88 more)
- gps_accuracy_poor: 8632 row(s) affected
- quaternion_invalid: 431 row(s) affected
- attitude_out_of_bounds: 12 row(s) affected

norm_FLY002.csv

- missing_gps: 16 row(s) affected
- gps_poor_fix: 3164 row(s) affected
- contains_no_value: 19 column(s) - 4.3.0, ConvertDatV3, IMUCalcs(0):Lat:C, IMUCalcs(0):Long:C, IMUCalcs(0):diffVelD:C (+14 more)
- contains_constant_value: 90 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+85 more)
- gps_accuracy_poor: 2338 row(s) affected
- quaternion_invalid: 120 row(s) affected
- attitude_out_of_bounds: 592 row(s) affected

norm_FLY003.csv

- missing_gps: 31 row(s) affected
- gps_poor_fix: 536 row(s) affected
- contains_no_value: 22 column(s) - 4.3.0, ConvertDatV3, IMUCalcs(0):Lat:C, IMUCalcs(0):Long:C, IMUCalcs(0):diffVelD:C (+17 more)
- contains_constant_value: 93 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+88 more)
- gps_accuracy_poor: 981 row(s) affected
- quaternion_invalid: 204 row(s) affected

norm_FLY004.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 11 row(s) affected
- zero_coordinate: 251 row(s) affected
- gps_poor_fix: 251 row(s) affected
- contains_no_value: 27 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+22 more)
- contains_constant_value: 95 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+90 more)
- gps_accuracy_poor: 251 row(s) affected
- quaternion_invalid: 253 row(s) affected

norm_FLY005.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 6543 row(s) affected
- gps_poor_fix: 16303 row(s) affected
- contains_no_value: 19 column(s) - 4.3.0, ConvertDatV3, IMUCalcs(0):Lat:C, IMUCalcs(0):Long:C, IMUCalcs(0):diffVelD:C (+14 more)
- contains_constant_value: 90 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+85 more)
- gps_accuracy_poor: 16477 row(s) affected
- quaternion_invalid: 118 row(s) affected

norm_FLY006.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 827 row(s) affected
- altitude_negative: 1545 row(s) affected
- gps_poor_fix: 827 row(s) affected
- contains_no_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost (+1 more)
- contains_constant_value: 49 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D,

BatteryInfo:out_ctl:D (+44 more)

- gps_accuracy_poor: 827 row(s) affected
- quaternion_invalid: 263 row(s) affected
- attitude_out_of_bounds: 69 row(s) affected

norm_FLY007.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 31 row(s) affected
- zero_coordinate: 2 row(s) affected
- altitude_spike: 1 row(s) affected
- motor_airborne_off: 142 row(s) affected
- gps_poor_fix: 16 row(s) affected
- contains_no_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost
- contains_constant_value: 43 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+38 more)
- gps_accuracy_poor: 16 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY008.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 4060 row(s) affected
- gps_poor_fix: 4060 row(s) affected
- contains_no_value: 27 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+22 more)
- contains_constant_value: 100 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+95 more)
- gps_accuracy_poor: 4060 row(s) affected
- quaternion_invalid: 768 row(s) affected

norm_FLY009.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 97577 row(s) affected
- gps_poor_fix: 97577 row(s) affected
- contains_no_value: 32 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+27 more)
- contains_constant_value: 107 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+102 more)
- gps_accuracy_poor: 97577 row(s) affected
- quaternion_invalid: 150 row(s) affected

norm_FLY010.csv

- timestamp_gap: 3 row(s) affected
- missing_gps: 7 row(s) affected
- zero_coordinate: 16287 row(s) affected
- gps_poor_fix: 57233 row(s) affected
- contains_no_value: 32 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+27 more)
- contains_constant_value: 107 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+102 more)
- gps_accuracy_poor: 57233 row(s) affected

norm_FLY011.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 11 row(s) affected
- zero_coordinate: 101 row(s) affected
- gps_poor_fix: 101 row(s) affected
- contains_no_value: 59 column(s) - 4.3.0, ATTI_MINI0:pitch:C, ATTI_MINI0:roll:C, ATTI_MINI0:yaw:C, ConvertDatV3 (+54 more)
- contains_constant_value: 136 column(s) - ATTI_MINI0:s_agz0, ATTI_MINI0:s_cnt0, ATTI_MINI0:s_pgz0, ATTI_MINI0:s_qw0, ATTI_MINI0:s_qx0 (+131 more)
- gps_accuracy_poor: 101 row(s) affected
- quaternion_invalid: 106 row(s) affected

norm_FLY012.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 11 row(s) affected
- zero_coordinate: 5719 row(s) affected
- gps_poor_fix: 5719 row(s) affected
- contains_no_value: 28 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+23 more)
- contains_constant_value: 99 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+94 more)
- gps_accuracy_poor: 5719 row(s) affected
- quaternion_invalid: 117 row(s) affected
- attitude_out_of_bounds: 125 row(s) affected

norm_FLY013.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 2836 row(s) affected
- gps_poor_fix: 2836 row(s) affected
- contains_no_value: 33 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+28 more)
- contains_constant_value: 107 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+102 more)
- gps_accuracy_poor: 2836 row(s) affected
- quaternion_invalid: 153 row(s) affected

norm_FLY014.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 11 row(s) affected
- zero_coordinate: 3330 row(s) affected
- gps_poor_fix: 3330 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 88 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+83 more)
- gps_accuracy_poor: 3330 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY015.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 3923 row(s) affected
- gps_poor_fix: 7527 row(s) affected
- contains_no_value: 33 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+28 more)
- contains_constant_value: 105 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+100 more)
- gps_accuracy_poor: 7527 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY016.csv

- missing_gps: 9 row(s) affected
- zero_coordinate: 19517 row(s) affected
- gps_poor_fix: 19517 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 104 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+99 more)
- gps_accuracy_poor: 19517 row(s) affected
- quaternion_invalid: 1 row(s) affected

Flight Logs**2018-06-19_17_00_39-08RDD9100100KG.dat**

- format: json_array_log | entries: 6

upgradeLog_0.txt

- format: bracketed_log | entries: 1861

Flight Records**norm_DJIFlightRecord_2018-06-19_17-00-38.csv**

- contains_no_value: 40 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, APP_SER_WARN.warn, CENTER_BATTERY.connStatus (+35 more)
- contains_constant_value: 172 column(s) - APP_TIP.tip, CAMERA_INFO.cameraType, CAMERA_INFO.connectState, CAMERA_INFO.enabledPhoto, CAMERA_INFO.encryptStatus (+167 more)

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Drone Model inconsistency detected - conflicting values observed: Mavic Pro, MavicPro, UNKNOWN. Manual verification required.
2. Note: no firmware version found - Device and account information.
3. Note: no last app launch found - Device and account information.
4. 11 flight record(s) could not be correlated to a drone log (flight_01, flight_02, flight_03, flight_04, flight_05, flight_06, flight_07, flight_08, flight_09, flight_10, flight_11). Independent corroboration is absent for these flight(s).
5. 7 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

com.dji.go.plist		Account Data	controller_ios	extract_logical
SHA-256:	cfc595c322da56f52b65e7b181f78fc964ee20585129213c5bd05a0bbb5fa38c			
SHA-1:	a8f7ccd5650092c9102c3ef97b1c79f29241e485			

5255DF85724869CD36D4AFCD12AE092D.db		Databases	controller_ios	extract_logical
SHA-256:	aae50532bee18355508de941e178ea847cd725018bb03a93589944f7cad4898e			
SHA-1:	7ecb7ed8a4b0ecf3028259a0eb0dcee998fcbc0c			

538103E3AB6E8F987EC10913E64A3E24.db		Databases	controller_ios	extract_logical
SHA-256:	6df74165eb44a8570bdc89f0038e0831de6ba509aa2aa2bd15aa1a6a2786cea0			
SHA-1:	f2d9e8167faa28bbb2685b330a7fabe2732943aa			

collectionList.db		Databases	controller_ios	extract_logical
SHA-256:	52ff80590e7a6e418b7aaeade8b04a62f661d77566de2e15827308fd1099cdb5			
SHA-1:	6ed495c00b14c89a44297e47ca17df469c6bcc2b			

djiFMDB.db		Databases	controller_ios	extract_logical
SHA-256:	5fdae73b94d5e7a078a8bfd1afda762eae9a5bff869139e9242dfec749fd52a			
SHA-1:	b1d200f2a2416e3c776ebc4e1779312198adf52b			

dji_fs_upgrade_local_cache.db		Databases	controller_ios	extract_logical
SHA-256:	94d2218b4ebd9a3d1590e861c319569224186adb86b14ac0ce78084152af9efc			
SHA-1:	2ab8f305677bf2f5333b49b1bd8ff66e14e01009			

flysafe_app_dynamic_areas.db		Databases	controller_ios	extract_logical
SHA-256:	1ddddd598a533c2a733b45485b6c1c9334f31b7717c1fc3989eaa5ac3b95d6f1e			
SHA-1:	af91b109a173d5464fd932e7f7b0a500a1ea0a96			

flysafe_areas_djigo_en.db		Databases	controller_ios	extract_logical
SHA-256:	1a1c024072fba37bfbfbf1198f4227fcf2f4ffad347ad9a93b164a7077cd0b6f8			
SHA-1:	761bad6393f8f8a28ecea57069626b9a9eaa6de4			

flysafe_dji_flight_dynamic_areas.db		Databases	controller_ios	extract_logical
SHA-256:	4ca303a13618ffcd042aa7200918efa332709c2997d60cf183fee20881794e4f			
SHA-1:	4788cd3336d179bb64f2ed63bf1eadc02b7eb964			

flysafe_license_unlock.db		Databases	controller_ios	extract_logical
SHA-256:	f720dc5f6a98d9e4c46a1cfb62f18614c68d495ae942fe05ad3528c327acac5f			
SHA-1:	48f3ac4943447b209aa61346da57c4096c444dba			

flysafe_static_circle.db		Databases	controller_ios	extract_logical
SHA-256:	4787fc859c865dab6b3d0db8eb68a55c764ca5d13105b14429cc2d455708cfb4			
SHA-1:	ccf2d5995e798abe1f1255c252c28ec8ae699dd4			

2018-06-19_15-42-28_FLY057.DAT		Drone Logs	controller_ios	extract_logical
SHA-256:	f671665aa3674312f39bb013b0b9919d041ead15aacc775ab2c1611de9ba7d0c			
SHA-1:	096b559c1954007ef987d6e329704ea1600df0ae			

2018-06-19_15-46-44_FLY058.DAT		Drone Logs	controller_ios	extract_logical
SHA-256:	83149907babd2d9d37767b0242b38addabe0f46ed5e909ea7071cf06261ea24b			
SHA-1:	565dabd2f03cb341b2415e7055ce49694fc37b40			

2018-06-19_15-47-44_FLY058.DAT		Drone Logs	controller_ios	extract_logical
SHA-256:	936aeec7efdf5f30e06ca3c11d544b2c51bc09fe62337cfee510e90535d1d17f			
SHA-1:	1c9519717c0740f2e6939da6484a0140ea487d2c			

2018-06-19_16-31-40_FLY059.DAT		Drone Logs	controller_ios	extract_logical
SHA-256:	6a5361c6bf3390e457d5f1c2f6d4b0beb5ee6fba11bb4ef9614bfbb13233a2			
SHA-1:	d9cb3a3b8f03c38a9f175f031f549c17db0ba818			

2018-06-19_16-45-50_FLY060.DAT		Drone Logs	controller_ios	extract_logical
SHA-256:	b5aebbf9a90b5c22d19b25b52dc02c643e036abc3535ada1c8e2f497e8f7ffd0			
SHA-1:	0d1de58f3a311a5924030f45d14439181db6f4ae			

2018-06-19_16-56-14_FLY061.DAT		Drone Logs	controller_ios	extract_logical
SHA-256:	ad33208ee9a55ec3969317b076fb205e1de1f82b2ea6d58cfbdd4685780812e4			
SHA-1:	0239bc4282e8b4e523fc618cb4ad30115a5fe0dc			

FLY000.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	1a9879a8701812f7db9be3ca2e5353c13acc70d49364a7ebe9c927eebeb15d05			
SHA-1:	ce3d13365a58fdf4c372e844e6f55c3ef32995c4			

FLY001.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	8384e5f02578cd2be75ff9c9992dcd93c8d648f5442b86b5eccacdd0c57f3b08			
SHA-1:	b139ef88cdd114c3ff7ed3a2725576a5d39a5f72			

FLY002.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	c57b8274a24441836fef958c18603ee46ad080177b6dee29d1f169cda27c773c			
SHA-1:	3bcf40ce6380733b4aacf2a5b6735c3ab6752ca9			

FLY003.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	bff8848556bd8e3039ce7ce7e037ba9d7831bdf9f51abc4bda54fc2a4706dba			
SHA-1:	433ff6b9d589c2d159329d9798b879db085877bb			

FLY004.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	a76d8a13c025eba285a816d2e3e2b789ebdd96deb6d86813da1c6e9756d3f588			
SHA-1:	7838b88c81ca399e40909ee94037b6cc8b6fa965			

FLY005.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	19a181563c1079aa12ee9ab2a9526e29b213acdc7cd604fafb7b744e8e8a8b85			
SHA-1:	e827485ed85883a175cb3b01e52242be1e362950			

FLY006.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	ce98b39611431b8818967a19fdf9748517f98fae36948b29e7d7ca2a068288d8			
SHA-1:	a508090b00a75fc86eeb2361ad5369f59312be47			

FLY007.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	3224fdc41ec66e306605099e263516ce515fb2c539477d4b18fbabdfc9b0ad50			
SHA-1:	5f2faa920479e3fde70312d4b4824455ff392e5e			

FLY008.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	abd42bc17f89951f7b22f87a0255179e12ed9fbeb71b991ce5b2090a5d991337			
SHA-1:	f4d29a9931b5f91f164c56228b720414ef6a2d22			

FLY009.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	5bb67278147aeba97c73e10a7a2bc60426e3b5b75bca21478ae3fd39ee849086			
SHA-1:	a58efecb2298955d5bf12e93e15044f459d4252			

FLY010.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	08ffdc892dd04d1c67005a8573597fd8d70a5f3885c6b7ee5c627dad7f29f185			
SHA-1:	64bec2fece57b1b2c2f240918ec4e490ce87d9cd			

FLY011.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	464c843c1069e1ec77e2fb152c0c0e15586f51fecf2912ad3cb7b5a8b87bd54f			
SHA-1:	34306cd094bbc7dc4553a962bc48bfcd4c00893			

FLY012.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	7f733632ee36c97ebaa57309bc4ab22a97c168e6c10b83bca0650bd569d6ff62			
SHA-1:	22468ca1a41bbf7a5372b66cc1d6a8973010d350			

FLY013.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	d9fdb5331a809593ddea2481ad4345ee5fa110db40b385a12695d7bc81e6908c			
SHA-1:	44e3700af623cb6c643fc5c26e81b2507ac1b790			

FLY014.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	a5874b46a26ca22f6e3818fc313a47b390ef98449932da285f26ca9ba6469ea5			
SHA-1:	0b1f56ebaac496f27c415deea1defacb49a3f544			

FLY015.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	39b992905ea24f7d15f2a19068e2da57772702222ea0b59f544eeac51caa2850			
SHA-1:	348d82d8a5076875c1a17841b49e89222a18d50e			

FLY016.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	6a29b6b33b090c188a1041a660a12a8c91353b16ab5771e5a91f98ac83c16b3c			
SHA-1:	28f96409d1cf3c06515d84baf355a63c4b863a32			

2018-06-19_17_00_39-08RDD9100100KG.dat		Flight Logs	controller_ios	extract_logical
SHA-256:	dc6ea1c3002ee63fbfb3cc3bb23389853a040647beea0ab3622a0d0c915af2ac			
SHA-1:	059aef4cf18112873acbe4a3f157d0ce7b5e4284			

Mavic_Pro_1301_upgrade_0.dat		Flight Logs	controller_ios	extract_logical
SHA-256:	5f2ef88afb76eea173d78b8daf42462e61bc0fcda766938b29c08d1224f60c75			
SHA-1:	b920039de08a28b550a4c1eeb6f1d8bc9fc9b6e0			

upgradeLog_0.txt		Flight Logs	controller_ios	extract_logical
SHA-256:	f1ea36a4728c1e3703990cbb3cba3cd122a8119e220a8f98d429cd767361fa20			
SHA-1:	d51a619cef48660ac88e99983c897c582f6175d0			

DJIFlightRecord_2018-06-19_17-00-38_.txt		Flight Records	controller_ios	extract_logical
SHA-256:	24a6826c37373e1c1f02b410f8477c19a6c6c5cf8acbc781725a5b2f5dffffa65			
SHA-1:	a8ed0ab03690245b6bcbcd6e26fe9bda3d249a2a			

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Ios

Category	P3 Extracted	P4 Decoded	P5 Normalised
Account Data	1	1	0
Databases	10	0	0
Drone Logs	6	12	6
Flight Logs	3	0	0
Flight Records	1	1	1

Drone Flight Storage

Category	P3 Extracted	P4 Decoded	P5 Normalised
Drone Logs	17	34	17

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-18 11:45:46 mmls vThe Sleuth Kit ver 4.15.0 rc=1

cmd: mmls.exe df019_internal.001

2026-06-18 11:45:46 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p df019_internal.001

2026-06-18 11:46:43 parser_ios v- rc=n/a

cmd: ios_logical.zip C:\Users\Floris\Documents\dfdof_output\ft-df019-2018-ios\p2_image_parsing\controller_ios_parsed
out: controller_ios_parsed

2026-06-18 11:46:44 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_internal.001 5
out: FLY000.DAT

2026-06-18 11:46:44 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_internal.001 6
out: FLY001.DAT

2026-06-18 11:46:45 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_internal.001 8
out: FLY002.DAT

2026-06-18 11:46:45 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_internal.001 9
out: FLY003.DAT

2026-06-18 11:46:45 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_internal.001 10
out: FLY004.DAT

2026-06-18 11:46:46 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_internal.001 11
out: FLY005.DAT

2026-06-18 11:46:46 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_internal.001 12
out: FLY006.DAT

2026-06-18 11:46:47 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_internal.001 13
out: FLY007.DAT

2026-06-18 11:46:47 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_internal.001 14
out: FLY008.DAT

2026-06-18 11:46:49 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_internal.001 15
out: FLY009.DAT

2026-06-18 11:46:51 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_internal.001 16
out: FLY010.DAT

2026-06-18 11:46:52 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_internal.001 17
out: FLY011.DAT

2026-06-18 11:46:52 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_internal.001 18

out: FLY012.DAT

2026-06-18 11:46:52 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_internal.001 19

out: FLY013.DAT

2026-06-18 11:46:52 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_internal.001 20

out: FLY014.DAT

2026-06-18 11:46:52 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_internal.001 21

out: FLY015.DAT

2026-06-18 11:46:53 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_internal.001 22

out: FLY016.DAT

2026-06-18 11:47:22 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe 2018-06-19_15-42-28_FLY057.DAT

out: 2018-06-19_15-42-28_FLY057.csv, 2018-06-19_15-42-28_FLY057.kml

2026-06-18 11:47:36 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe 2018-06-19_15-46-44_FLY058.DAT

out: 2018-06-19_15-46-44_FLY058.csv, 2018-06-19_15-46-44_FLY058.kml

2026-06-18 11:47:52 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe 2018-06-19_15-47-44_FLY058.DAT

out: 2018-06-19_15-47-44_FLY058.csv, 2018-06-19_15-47-44_FLY058.kml

2026-06-18 11:48:22 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe 2018-06-19_16-31-40_FLY059.DAT

out: 2018-06-19_16-31-40_FLY059.csv, 2018-06-19_16-31-40_FLY059.kml

2026-06-18 11:48:37 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe 2018-06-19_16-45-50_FLY060.DAT

out: 2018-06-19_16-45-50_FLY060.csv, 2018-06-19_16-45-50_FLY060.kml

2026-06-18 11:48:56 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe 2018-06-19_16-56-14_FLY061.DAT

out: 2018-06-19_16-56-14_FLY061.csv, 2018-06-19_16-56-14_FLY061.kml

2026-06-18 11:48:57 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2018-06-19__17-00-38_.txt DJIFlightRecord_2018-06-19__17-00-38_.csv

out: DJIFlightRecord_2018-06-19__17-00-38_.csv

2026-06-18 11:49:22 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY000.DAT

out: FLY000.csv, FLY000.kml

2026-06-18 11:49:36 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY001.DAT

out: FLY001.csv, FLY001.kml

2026-06-18 11:49:51 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY002.DAT

out: FLY002.csv, FLY002.kml

2026-06-18 11:50:07 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY003.DAT

out: FLY003.csv, FLY003.kml

2026-06-18 11:50:21 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY004.DAT

out: FLY004.csv, FLY004.kml

2026-06-18 11:50:36 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY005.DAT

out: FLY005.csv, FLY005.kml

2026-06-18 11:50:53 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY006.DAT

out: FLY006.csv, FLY006.kml

2026-06-18 11:51:14 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY007.DAT

out: FLY007.csv, FLY007.kml

2026-06-18 11:51:32 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY008.DAT

out: FLY008.csv, FLY008.kml

2026-06-18 11:51:50 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY009.DAT

out: FLY009.csv, FLY009.kml

2026-06-18 11:52:08 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY010.DAT

out: FLY010.csv, FLY010.kml

2026-06-18 11:52:25 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY011.DAT

out: FLY011.csv, FLY011.kml

2026-06-18 11:52:40 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY012.DAT

out: FLY012.csv, FLY012.kml

2026-06-18 11:52:59 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY013.DAT

out: FLY013.csv, FLY013.kml

2026-06-18 11:53:15 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY014.DAT

out: FLY014.csv, FLY014.kml

2026-06-18 11:53:28 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY015.DAT

out: FLY015.csv, FLY015.kml

2026-06-18 11:53:42 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY016.DAT

out: FLY016.csv, FLY016.kml

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap ≥ 60 s AND $\geq 90\%$ of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs ≤ 25 m with ≥ 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a ± 2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.